Volatility and Under-Insurance in Economies with Limited Pledgeability: Evidence from a Frost Shock*

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Abstract: We examine how a physical climate shock propagates through collateral constraints and supply chains, using administrative microdata on bank credit, interfirm payments, and crop insurance around Brazil's July 2021 frost. The frost destroyed coffee trees, a perennial crop that we model as capital used to secure farmers' borrowing, so the frost reduced pledgeable collateral in addition to current cash flow. Crop insurance premiums in Brazil average 30% below actuarially fair levels, yet only 20% of coffee farmers were insured. After two growing seasons, each percentage point of frost damage led to 1.4% less debt and 0.38% less paid to upstream input suppliers among uninsured farmers, while input purchases increased by 0.24% among insured farmers. Default and interest rates were unchanged. Instead, debt adjusted via forbearance and quantities. We develop a general equilibrium model of climate risk under limited commitment in which borrowing is secured by future harvest revenue and by land. Limited pledgeability explains why constrained farmers forgo subsidized insurance ex ante, since the upfront premium consumes the debt capacity that finances production, and why post-disaster emergency credit subject to the same collateral constraint fails to reach damaged farmers: only 49% of Brazil's frost credit line was disbursed, and 40% of the lending went to the insured 20%. The endogenous coffee price restores collateral value only for surviving capital, so the change in borrowing capacity depends on which asset the farmer pledges. Supervisory climate stress tests that interpret low bank losses as resilience miss the risk retained by farmers, their suppliers, and consumers.

Keywords: Physical Climate Risk; Insurance Protection Gap; Collateral Constraints; Agricultural Credit; Supply Chains; Climate Stress Testing.

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1 Introduction

Around the world, two policies characterize physical climate risk management: subsidized insurance before a disaster and emergency credit afterwards. Despite government intervention, both policies leave risk gaps that we model and show empirical evidence to justify. Crop insurance subsidies exceed $20 billion globally, yet under-insurance is pervasive (Hazell and Varangis, 2020; Kramer et al., 2022), and the gap appears in developed markets too (Sastry et al., 2024). Post-disaster credit frequently fails to reach those who sustained the worst damage. This paper studies why these two adaptation mechanisms break down together, and how a physical climate shock propagates through credit markets and production networks. Much of the existing literature either seeks behavioral explanations for low take-up or estimates insurance demand in reduced form. We show instead that under-insurance, the failure of post-disaster credit, and the supply-chain propagation of climate shocks share one explanation: limited collateral pledgeability. Insurance and production compete for the same borrowing capacity, so the producers with the most to insure are the least able to pay for coverage in advance or to rebuild afterward.

We study the aftermath of Brazil's July 2021 frost, an extreme weather event that destroyed productive capital across Brazil's main coffee-growing states. Coffee is a perennial crop whose trees take several years to bear fruit. For coffee farmers, coffee trees are a core component of farm capital and pledgeable collateral, so the frost reduced collateral value in addition to current output. We combine transaction-level administrative data from the Central Bank of Brazil: the credit registry (SCR), interfirm payment records, the rural credit registry with its linked public crop insurance program (SICOR), and data from the Ministry of Agriculture (MAPA) on contracts from the main private crop insurers. We are the first to link credit, insurance, and firm-to-firm payment microdata for the same farmers, including small farmers who borrow under a personal identifier rather than a firm identifier. Among uninsured farmers, each percentage point of frost damage led to 1.4 percentage points less debt and 0.38 percentage points less paid to upstream suppliers. Greater frost severity reduced subsequent take-up of insurance despite the increased salience, and despite community-rated premiums and interest rate that remained unchanged after the shock. Among insured farmers, input purchases that we interpret as recovery investment increased by 0.24 percentage points per percentage point of frost damage. Our findings are inconsistent with models that treat credit and insurance as substitutes. Instead, our findings are consistent with a constrained environment where borrowing capacity and insurance interact dynamically through collateral constraints.

To interpret these empirical patterns and trace their aggregate implications, we develop a general equilibrium model with a production economy subject to climate shocks under limited commitment, building on Rampini and Viswanathan (2022). Risk-averse farmers face shocks to their capital stock that reduce both output and pledgeable collateral. The core frictions are that farmers can divert part of current output and lenders can seize only part of land value, so only a fraction of future value can be pledged. Markets are endogenously incomplete. Insurance reallocates resources across states, but the premium must be paid up front, using borrowing capacity that could otherwise finance production. Under-insurance is a constrained-optimal choice: as in Rampini and Viswanathan (2022), low-net-worth farmers use all available financial slack and do not purchase insurance in order to preserve their capacity to borrow for operational and personal expenses. After a shock, these farmers can have both the highest marginal return to capital and the least ability to finance their rebuilding.

We extend this framework by allowing farmers to borrow against two types of collateral. First, a *Cédula de Produto Rural* (CPR) contract pledges a share of future harvest revenue. Second, farmers can pledge land when property rights and enforcement give the lender a recoverable claim against its value. We separate land value into the value of bare land, which is not directly affected by the frost, and the value of the standing coffee grove. The frost reduces the future harvest and destroys part of the grove, but it also increases coffee prices. The same event that destroys the asset raises the price of what survives, so the map from shock to collateral value is non-monotonic, motivating our general equilibrium structure.

Our main theoretical contribution is endogenizing the coffee price. Aggregate supply determines the output price, which enters both farmers' revenue and the value that they can pledge. The price increasedsa works through two channels: a budget channel that raises revenue on surviving output, and a collateral channel that raises the pledgeable value of surviving harvest contracts and groves. The loss of a farmer's own collateral is distinct from the effect of the common coffee price on the borrowing constraints of other farmers, which is what generates collateral externalities in Dávila and Korinek (2018). This distinction allows the model to separate the direct effect of the frost from the general equilibrium collateral effect, and it implies that an aggregate shock widens the gap between damaged and undamaged farmers, rather than only shifting the level of borrowing. In the partial-equilibrium benchmark, we show that the optimal state-contingent contract can be implemented through endogenous borrowing limits, similarly to Alvarez and Jermann (2000), but without permanent exclusion from credit markets after default.

The model yields two implications for climate policy. The first implication is that post-disaster credit cannot substitute for insurance when it is intermediated under the same collateral constraint. Following the frost, the Central Bank of Brazil allocated 1.3 billion

BRL (approximately 250 million USD) to be disbursed as emergency credit lines to affected coffee farmers. Only 49 percent was lent, and 40 percent of the lending went to the 20 percent of farmers who already held insurance. In the model, an announced credit line does not expand the borrowing capacity of a farmer whose collateral has been destroyed. We consider two methods to relax the constraint. One is public guarantees that remove part of the promised repayment from the farmer's collateral constraint. Another is registries or monitoring that raise the pledgeability of harvest proceeds and land.

The second implication is that the framework changes how to interpret a climate stress test. Supervisory exercises, including those coordinated through the Network for Greening the Financial System, assess climate risk primarily through direct banking-sector losses. In our setting, default and interest rates did not move, consistent with lenders anticipating state-contingent repayment and granting maturity extensions. Low measured bank exposure did not indicate resilience. The risk stayed with farmers, who cut investment, and with their suppliers, who lost sales. Under the model's inelastic-demand benchmark, the reduction in aggregate supply passes part of the loss to consumers through the coffee price. A stress test that monitors only financial-sector solvency might miss how the real sector absorbs the shock, which we show in theory and empirics.

Our model contributes to the theoretical financial literature on the causes and consequences of under-insurance. **?** establish that hedging matters because investment and financing policies must be coordinated when external finance is costly. Caballero and Krishnamurthy (2003) show that financially constrained firms undervalue insurance when the marginal value of liquidity in production exceeds the market price. Dávila and Korinek (2018) show how financial frictions generate distributive and collateral externalities. Martins-Da-Rocha et al. (2022) emphasize the role of limited pledgeability and strategic default. Unlike models where amplification operates through the interest rate, our mechanism operates with a fixed funding rate through changes in the output price and its effect on collateral. Sastry et al. (2024) document a comparable protection gap in US property insurance and provide evidence consistent with the Rampini and Viswanathan (2022) mechanism, but focus on household insurance demand. In contrast, we study firms' borrowing and investment responses to a realized shock to productive capital and collateral.

More broadly, our paper adds to a literature documenting that financial frictions amplify the real effects of shocks through lower investment, distorted insurance decisions, and endogenous price dynamics. Alfaro et al. (2024) show that financial frictions more than double the impact of aggregate uncertainty shocks on investment and financial variables, whereas our mechanism turns on the interaction of capital destruction, collateral constraints, and price feedback in general equilibrium. Other papers, such as Arellano et al. (2019), Ottonello and Winberry (2020), and Giroud and Mueller (2017) show how financial frictions propagate shocks through investment, credit, and labor channels, but do not study the role of hedging. Our model and regressions show that constrained under-insurance can amplify real responses through both extensive and intensive margins of credit and investment.

The model implies the atypical empirical specification and rationalizes four empirical findings. We estimate event studies on credit, investment, and insurance outcomes, comparing more and less affected farmers within municipality and initial insurance status. First, uninsured farmers experience larger declines in borrowing after the shock than insured farmers: among uninsured coffee farmers, experiencing a 1 percentage point more severe shock is associated with 1.4 percent less debt in the two years after the shock. Second, uninsured farmers reduce investment more, with each percentage point of shock severity associated with 0.38 percent less expenditure on upstream inputs used to rebuild capital, while affected insured farmers instead increase input purchases by 0.24 percent. Third, banks do not experience higher default rates and grant forbearance instead, extending outstanding debt maturity by 0.2 months per percentage point of damage, roughly five months at the average damage of 27 percent. Fourth, among farmers insured at the time of the frost, those who experience a more severe shock are less likely than less affected farmers nearby to subsequently purchase insurance, even though insurance pricing is community-rated. By comparison, there is no relationship between insurance take-up and frost shock severity for farmers of annual crops such as corn and wheat, which are replanted each season. Unlike models with endogenous exclusion or interest rate spreads, our framework generates under-insurance through collateral limits in a setting where borrowers are not excluded after default and loan rates are largely administered.

In addition to the contribution of the model linked to the empirics, this paper contributes to two empirical strands of the literature. The first is an empirical literature on the interaction between credit and insurance. One strand focuses on the role of ex ante financial constraints that can be alleviated by financing the insurance premiums (Cole and Xiong, 2017; Casaburi and Willis, 2018). In our setting, subsidized lending is linked to subsidized insurance premiums, which McIntosh et al. (2013) show increases willingness to pay for insurance but Annan (2022) show can exacerbate moral hazard. Yet insurance take-up is still low, so we believe that financing premiums is not a sufficient explanation. Lane (2024) follow a long literature that studies whether credit can substitute for insurance. We show that credit and insurance are complements after a shock despite appearing substitutable ex ante, and we show how this arises from collateral constraints that bind ex post rather than ex ante. We are also the first to construct a comprehensive dataset linking credit and insurance from both formal (firm) and informal (individual) contracts with banks and insurers at the transaction level.

The second is a long empirical literature in corporate finance, with a burgeoning literature in climate finance, on the impact of weather shocks on farmers through financial channels. Bergman et al. (2020) estimate the elasticities of land prices and revenue to yields using a weather shock instrument, in the setting of the farm debt crisis in the 1980s in Iowa before the introduction of crop insurance. Brown et al. (2021) estimate how firms respond to severe weather shocks using credit lines, and find that heavier snow at a firm's headquarters decreases cash flow and increases short-term lending with no impact on investment nor capital stock. Cortés and Strahan (2017) show that there is reallocation of lending across locations in response to a natural disaster shock as banks' balance sheets deteriorate. Benincasa et al. (2024) document how firms finance losses from extreme weather events, and Giroud et al. (2012) use exogenous weather variation to show how leverage constrains investment. Weagley (2019) shows that intermediary capacity determines the price of weather risk transfer, which is the supply-side counterpart to the demand-side constraint that we study. Deschênes and Greenstone (2007) estimate the effect of weather on agricultural output without the financial channel that is central here. In our setting, there is a dominant nationwide agricultural sector lender backstopped by the government, so lending rates did not increase after the shock, allowing us a clean test of the insurance mechanism. In general, our setting and mechanism differ from each of these papers: the interaction of the physical shock, causing a reduction in the capital stock, with insurance generates our financial and hedging results.

Our setting is well-suited for our study for three reasons. The first reason is the comprehensive availability of insurance, combined with the common puzzle of low take-up despite subsidized premiums and insurance-linked financing for the premiums. In most non-agricultural settings, revenue insurance is unavailable. In agriculture in Brazil, every farmer can access public insurance, and most can choose to buy additional insurance from private providers. Premiums average 30 percent below actuarially fair levels, yet take-up for coffee is as low as 20 percent. The second reason is the detailed data available on farm production and assets that we merge with payments and credit microdata, allowing us to trace the domestic impacts of the shock at a granular level. Finally, agriculture experiences shocks large enough to affect aggregate output, but also unpredictably heterogeneous enough over nearby locations to provide quasi-exogenous variation for estimation. Identification comes from differences in frost severity across farmers in the same municipality with the same initial insurance status.

Netto (1959) was the first to describe the coffee cycle formally, using a theoretical model and historical data to show how expansion and crisis alternate because farmers respond to past prices and coffee plants take time to mature. Our model provides a different mechanism with forward-looking farmers: a frost raises coffee prices at the same time that it destroys productive capital, so the price increase relaxes borrowing constraints most for farmers whose capital and price-sensitive collateral survive, while affected constrained farmers may rebuild slowly. The model event study illustrates how this difference can make a one-time shock persistent.

The remainder of the paper is organized as follows. Section 2 presents the model and the model-generated event studies. Section 3 describes the institutional setting in Brazil, the data, and the timeline of the frost shock. Section 4 introduces the empirical methodology and presents the regression results. Section 5 considers policy implications for emergency credit, collateral pledgeability, and climate stress testing. Section 6 concludes.

2 Model

To interpret the empirical patterns and evaluate their aggregate implications, we first develop a general equilibrium model with capital shocks, borrowing constraints, and endogenous price determination. We build upon Rampini and Viswanathan (2013, 2022), but consider a production economy that experiences both aggregate and idiosyncratic capital stock shocks similar to Caballero and Krishnamurthy (2003) and Lorenzoni (2008). Farmers can borrow against two forms of collateral that are important in our setting. A CPR pledges future harvest revenue, while a land-secured loan pledges the value of bare land and, depending on the appraisal, the standing coffee grove.

The distinction between these forms of collateral matters after a frost. The frost destroys productive capital and reduces the future harvest, but does not destroy bare land. At the same time, the aggregate reduction in coffee supply raises the coffee price. The price increase raises current revenue and the value of the harvest and grove that survive the shock. The model therefore features both a budget channel, through the effect of price on farmers' internal funds, and a collateral channel, through the effect of price on pledgeable value. These channels can offset part of the aggregate loss while increasing differences across farmers with different physical damage and collateral.

There are three versions of the model within the recursive environment below. First, setting $\lambda = \chi = 0$, allowing direct recovery of productive capital, and taking the price process as given produces the capital-constraint partial-equilibrium model that is fully solved. We present its derivation and numerical results in Appendix C.5. Second, setting $\lambda_K = \chi = 0$ gives the CPR-only model with an output-based borrowing constraint. The fixed-price household problem is solved and produces the paired event studies below. Third, setting $\lambda_K = 0$ and allowing both $\lambda > 0$ and $\chi > 0$ gives the full model with CPR and land collateral. The single constraint below therefore nests all three versions. We feature the CPR and land version in the exposition because it describes the actual collateral used by farmers.

2.1 Environment and Timing

There is a continuum of risk-averse farmers indexed by $i \in [0,1]$. Farmer i has permanent productivity type $a \in \{L, H\}$, with productivity A_a, and preferences

$$U_i = \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u(c_{it}) \right], \qquad u(c) = \frac{c^{1-\sigma}}{1-\sigma}, \tag{1}$$

where $\beta \in (0,1)$ and $\sigma > 0$.

At the beginning of a period, a farmer has net worth W, aggregate state S, and individual state z. The farmer chooses consumption c, installed capital k, and a schedule of one-period state-contingent repayments $b(S', z')$. The aggregate state determines the incidence of frost damage, while the individual state determines the realization of the capital-survival rate $\theta(z', S')$.

The joint state follows a Markov process with transition probability $\pi(S', z' \mid S, z)$. Aggregate frost incidence determines the probability that an individual farmer receives the capital shock, so the model contains both aggregate and idiosyncratic risk.

Capital produces coffee after the next shock is realized:

$$y_a(S', z') = A_a \theta(z', S')^{\alpha} k^{\alpha}, \qquad \alpha \in (0,1), \tag{2}$$

and $\theta(z', S')k$ units of productive capital survive into the next period. Thus k is the pre-shock installed stock. In the quarterly computation, the normal-state value $\theta_n = 1 - \delta$ incorporates depreciation, while $\theta_f < \theta_n$ is the survival rate under frost.

Financial intermediaries are risk neutral, have deep pockets, and discount at the gross rate R, with $\beta R < 1$. Rural lending rates in our setting are largely administered, which motivates an exogenous R. Intermediaries can condition payments on verifiable shock realizations, but limited commitment restricts what they can recover following default. Farmers are not permanently excluded from future credit markets after default.

2.2 Limited Commitment, Debt, and Insurance

Let $d \geqslant 0$ denote non-state-contingent debt and let $h(S', z') \geqslant 0$ denote a state-contingent insurance payment. Define net repayment in state (S', z') by

$$b(S', z') \equiv d - h(S', z'). \tag{3}$$

Positive b is net debt repayment, while reducing b in a bad state provides insurance. Debt and insurance therefore determine repayment jointly in every state.

We write the solved benchmark and the current general equilibrium models using one enforcement constraint. Let $\lambda_K \in [0,1]$ be the recoverable share of surviving productive capital. This is the capital-based constraint in the solved benchmark. A CPR or harvest pledge makes fraction $\lambda \in [0,1]$ of next-period harvest revenue pledgeable, while fraction $\chi \in [0,1]$ of appraised land value can support repayment. The state-by-state borrowing constraint is

$$b(S',z') \leqslant \underbrace{\lambda_K \theta(z',S')k}_{\text{productive capital}} + \underbrace{\lambda p(S')A_a \theta(z',S')^{\alpha} k^{\alpha}}_{\text{CPR (harvest) pledge}} + \underbrace{\chi q(S',z')}_{\text{land pledge}}, \tag{4}$$

$$q(S',z') = \bar{L} + \psi G(S',z'), \qquad G(S',z') = \kappa_G p(S')\theta(z',S')k. \tag{5}$$

Here $\bar{L} \geqslant 0$ is the value of bare land, which is not damaged by the frost. The term G is the appraised value of the surviving grove. The parameter $\psi \in [0,1]$ is the share of grove value included in the appraisal, and κ_G converts surviving coffee capital into appraised value. We interpret χ as a reduced-form recovery rate that combines title quality, foreclosure, and enforcement. The current model does not track the physical transfer and subsequent ownership of land after default. We impose

$$\lambda_K + \chi\psi\kappa_G p(S') \leqslant 1 \quad \text{for every } S', \tag{6}$$

so that the combined claims against productive capital and grove value do not exceed the replacement cost of surviving capital. We include land in the collateral constraint but not in the flow budget because the same grove capital already produces output and survives into the next period. The variable θk already includes normal depreciation in the recursive model. It corresponds to $(1-\delta)\theta k$ in the notation of the solved benchmark.

We set $\lambda_K = 0$ when comparing the collateral used in the general equilibrium model. **Regime I** has $\chi = 0$ and uses only CPR collateral. **Regime II-a** has $\chi > 0$ and $\psi = 0$, so land provides a fixed collateral cushion. **Regime II-b** has $\chi > 0$ and $\psi > 0$, so land value depends on both grove damage and the coffee price. To give II-a and II-b the same normal-state land value, let G_n be grove value for a reference farm in the normal state and impose

$$\bar{L}^{\text{IIa}} = \bar{L}^{\text{IIb}} + \psi G_n. \tag{7}$$

Without this adjustment, the comparison would also change the initial level of collateral. In the heterogeneous model, we evaluate G_n at the pre-shock steady-state mean.

Given this enforcement limit, any state-contingent net repayment can be written as debt minus insurance.

Proposition 1 (Debt–Insurance Representation). Any bounded net-repayment schedule satisfying the enforcement limit can be implemented with one-period non-state-contingent debt d and nonnegative state-contingent insurance $h(S', z')$ satisfying

$$d \leqslant \lambda_K \theta(z', S')k + \lambda p(S')A_a \theta(z', S')^\alpha k^\alpha + \chi q(S', z') + h(S', z') \quad \text{for every } (S', z'). \tag{8}$$

Conversely, any such debt and insurance pair induces a net-repayment schedule satisfying the enforcement limit.

Proof. See Appendix C. The construction uses $b = d - h$ and the recovery value in (4). □

The proposition shows why credit and insurance can be complements. Insurance supports more debt by reducing repayment in bad states, but the farmer must pay the premium before the shock. A farmer with low net worth may therefore invest instead of buying insurance even when insurance is actuarially fair.

2.3 Recursive Equilibrium and Price Determination

Farmers take the perceived laws of motion for the distribution and price as given. We suppress the distribution in the notation: $V_a(W, S, z)$ denotes $V_a(W, S, z; \mu)$, and $p(S')$ denotes the perceived price $p(S', \mu')$. The farmer solves

$$V_a(W, S, z) = \max_{k, \{b(S', z')\}} \left\{ u(c) + \beta \mathbb{E}\left[V_a(W', S', z') \mid S, z \right] \right\} \tag{9}$$

$$\text{s.t.} \qquad c = W + \mathbb{E}\left[\frac{b(S', z')}{R} \mid S, z \right] - k, \tag{10}$$

$$W' = p(S')A_a \theta(z', S')^\alpha k^\alpha + \theta(z', S')k - b(S', z'), \tag{11}$$

$$b(S', z') \leqslant \lambda_K \theta(z', S')k + \lambda p(S')A_a \theta(z', S')^\alpha k^\alpha + \chi q(S', z') \quad \forall (S', z').$$

Net worth is the liquid state variable relevant for current choices. Insurance status is an endogenous outcome of the state-contingent repayment schedule rather than an exogenous type.

Let μ_t be the distribution of farmer type, wealth, and individual state before choices are made. Given the capital policy k_{it}, next period's output and price are

$$Y_{t+1} = \int_0^1 A_{a_i} \theta(z_{i,t+1}, S_{t+1})^\alpha k_{it}^\alpha \, di, \tag{12}$$

$$p_{t+1} = c_p Y_{t+1}^{-1/\gamma}, \qquad \gamma > 0. \tag{13}$$

The parameter γ is the quantity-demand elasticity. In equilibrium, farmer policies generate the distribution of installed capital, and the resulting output and price satisfy (12)–

(13).

A recursive competitive equilibrium consists of farmer value and policy functions, a law of motion for μ, and a price function such that: (i) farmer policies solve (9) subject to the budget and collateral constraints; (ii) intermediaries break even by pricing the repayment schedule at $\mathbb{E}[b(S', z')]/R$; (iii) the policy functions and shock process generate the next-period distribution; and (iv) aggregate output and the coffee price satisfy (12)–(13). In particular, equilibrium requires

$$\mu' = \Phi(\mu, S, S'; k_a, b_a), \qquad p(S', \mu') = c_p Y(S', \mu')^{-1/\gamma}, \tag{14}$$

where Φ is generated by the policy functions and the joint shock process. The equilibrium price is therefore an outcome of farmers' investment choices and the distribution of frost damage.

A widespread frost reduces output and raises the coffee price. The budget channel operates because the higher price raises revenue and next-period net worth for farmers with surviving production. The collateral channel operates because the same price enters both the CPR pledge and the grove component of land value. Holding capital fixed, the response of the borrowing limit to the price is

$$\frac{\partial \mathcal{B}_i(S', z'; k_i)}{\partial p(S')} = \lambda A_{a_i} \theta_i(S', z')^\alpha k_i^\alpha + \chi_i \psi_i \kappa_G \theta_i(S', z') k_i. \tag{15}$$

The price increase therefore relaxes borrowing constraints most for farmers with more surviving harvest and grove capital.

Let $\xi_i(S', z')$ be the multiplier on farmer i's collateral constraint. We summarize collateral-price exposure, holding policies and the distribution fixed, by

$$\Lambda^p(S') = \int \sum_{z'} \xi_i(S', z') \left[\lambda A_{a_i} \theta_i(S', z')^\alpha k_i^\alpha + \chi_i \psi_i \kappa_G \theta_i(S', z') k_i\right] d\mu(i). \tag{16}$$

Only farmers with binding constraints contribute to $\Lambda^p(S')$. It separates the direct effect of price on collateral from the effect of price on current revenue. The full equilibrium can therefore generate a partial aggregate offset at the same time that it widens the difference between affected farmers and farmers whose collateral survives.

2.4 Quantitative Solution Strategy and Calibration

To quantify the general equilibrium model, the solution strategy combines an endogenous grid method for the farmer's problem with a Krusell–Smith iteration for the price. Given a perceived law of motion for the price and each farmer type, the algorithm iterates on the marginal value of wealth. At each wealth point, it considers the unconstrained case, cases in which a subset of continuation-state constraints binds, and the fully constrained case. Because both damage and price vary across future states, it checks complementary slackness state by state rather than impose a fixed ranking from the best to the worst state. The resulting policy functions are interpolated over wealth and price.

The outer loop simulates the distribution of farmers, aggregates production using (12), and determines the price from (13). It updates the perceived law of motion by regressing next period's log price on current-state indicators, their interactions with the current log price, and next-state indicators. The household solution, simulation, and price update repeat until the price forecast and market-clearing residuals converge. This differs from the standard Krusell–Smith problem because future damage and price enter heterogeneous collateral constraints, the individual process combines aggregate and idiosyncratic shocks, and the output price is determined by inverse demand rather than a marginal product condition.

Table 1 reports the inputs for the four-state fixed-price CPR exercise and the starting values for the CPR and land general equilibrium models. They are inputs to the quantitative solution rather than estimates. The annual discount rate, funding rate, and depreciation rate are converted to the quarterly frequency in the recursive problem. These are separate from the parameter values used for the solved capital-constraint partial-equilibrium benchmark, which are reported in Table 4.

Table 1: Inputs for the Fixed-Price CPR Exercise

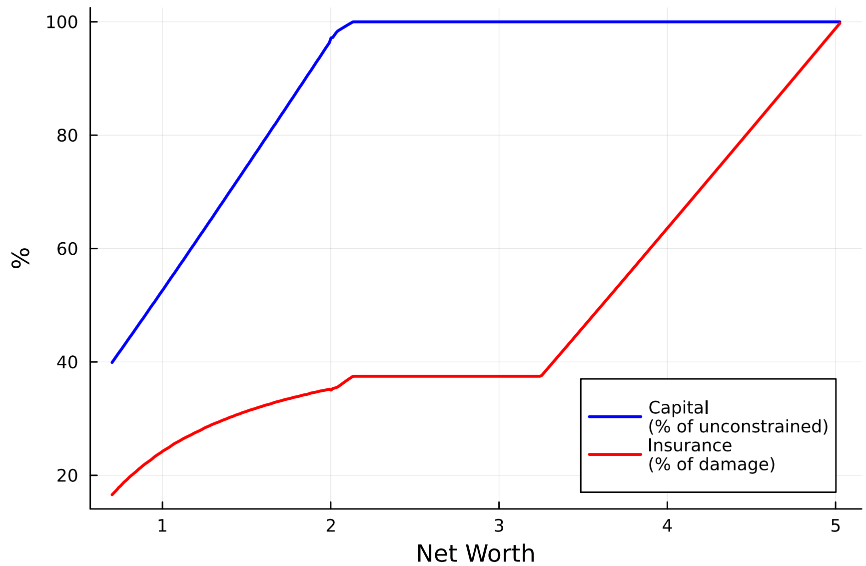
Parameter	Value	Interpretation
α	0.30	Capital elasticity
λ	0.40	Pledgeable share of harvest revenue
β^{ann}	0.85	Annual discount factor
R^{ann}	1.10	Annual gross funding rate
σ	2.00	CRRA coefficient
δ^{ann}	0.05	Annual normal depreciation
θ_f	0.30	Capital survival under frost
γ	0.40	Quantity-demand elasticity

At the quarterly frequency, we use

$$\beta = (0.85)^{1/4}, \qquad R = (1.10)^{1/4}, \qquad \delta = 1 - (0.95)^{1/4}. \tag{17}$$

The frost survival rate θ_f is event-specific and is not annualized. Figure 1 shows that our model replicates the Rampini and Viswanathan (2013, 2022) result on net worth and insurance. The lowest net worth farmers would not take up full insurance even under complete markets. Medium net worth farmers would be fully insured under complete markets, but are under-insured under incomplete markets.

Figure 1: Model-Implied Relationship Between Insurance and Net Worth



Notes: This figure shows the model-implied relationship between net worth and insurance coverage. Under incomplete markets, the binding pledgeability constraint results in lower insurance demand when net worth is low, compared to the complete market benchmark.

2.5 Model Event Study

The paired fixed-price CPR model event study uses four aggregate nodes that combine low or high frost incidence with a low or high coffee price.

Table 2: Aggregate States in the Model Event Study

State	Coffee Price	Probability of Frost Damage
s_1	1.0	s_n
s_2	1.5	s_n
s_3	1.0	s_f
s_4	1.5	s_f

Assume $s_f > s_n$, where f denotes frost and n denotes normal. Their unconditional distribution is $P_S = (0.1, 0.1, 0.6, 0.2)$. In the fixed-price exercise, the aggregate state is i.i.d., so each row of the transition matrix equals P_S. This four-node exercise separates farmers by initial insurance status. For comparison, Appendix C.5 reports the two-state, all-uninsured simulation and its parameters.[1]

We use the numerical solution of the fixed-price CPR household problem to construct the paired event study that motivates our empirical specification. For illustrative purposes, we assume $\lambda_K = \chi = 0$ and the four-node coffee-price process. We feed the model the following sequence of aggregate states:

$$S = \left[1, \underbrace{4}_{\text{frost and high price}}, \underbrace{2, 2}_{\text{high price}}\right]. \tag{18}$$

The first period is a normal state. In the second period, the economy enters the frost state, in which the probability of capital damage and the coffee price are both high. In the last two periods, the probability of damage returns to its normal level while the coffee price remains high.

The model generates a critical level of net worth, $\underline{W}$, below which optimal insurance covers less than 20% of the capital loss. We draw 1,000 farmers in each initial insurance group. Initial net worth is uniform on $[0, \underline{W}]$ for the uninsured group and on $[\underline{W}, \overline{W}]$ for the insured group. We then randomly assign half of the farmers in each group to receive the individual capital shock. Conditional on initial insurance status, the difference between shocked and unshocked farmers is therefore generated by the capital shock.

Let T denote the simulated frost date, let $\mathcal{T} = \{-1, 0, 1, 2\}$, and define

$$D_{it}^{\tau} = \mathbb{1}\{t - T = \tau\}\,\text{Shock}_i\,. \tag{19}$$

We estimate the following event study on the simulated panel:

$$y_{it} = \sum_{\tau \in \mathcal{T}\setminus\{-1\}} \beta_{\tau}^{I} D_{it}^{\tau}\,\text{Ins}_i + \sum_{\tau \in \mathcal{T}\setminus\{-1\}} \beta_{\tau}^{NI} D_{it}^{\tau}(1 - \text{Ins}_i) + \alpha_i + \alpha_{gt} + \varepsilon_{it}, \tag{20}$$

where g denotes initial insurance status. We normalize the period before the shock to zero and plot cumulative responses.

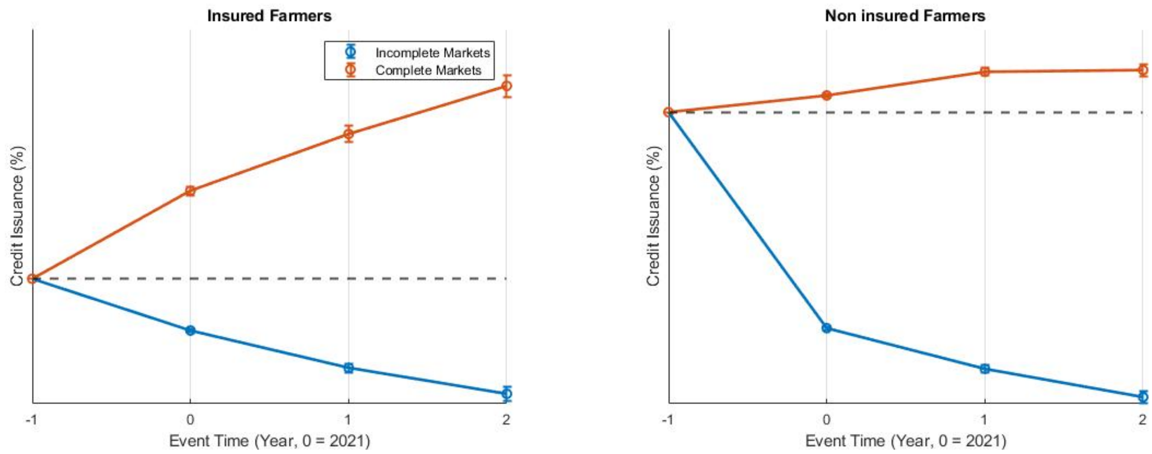
The first result is that insurance coverage increases with net worth. Under complete markets, farmers purchase full actuarially fair insurance. Under incomplete markets, farmers with low net worth use their available borrowing capacity to finance production and

[1] For the full GE estimation, the transition matrix $\pi(S' \mid S)$ will instead be disciplined by the historical frost process and optimal transport quantization. Conditional on validating the SCR guarantee fields, the parameters governing land collateral, $(\chi, \bar{L}, \psi, \kappa_G)$, will be disciplined by guarantee types, land-secured loan-to-value ratios, farmland prices, and grove values. The difference in borrowing responses between farmers whose pre-shock loans are predominantly secured by land and those whose loans are predominantly secured by harvest revenue will then enter as a moment for χ.

purchase less insurance.

Figure 2 shows the model-implied cumulative effects on borrowing. Uninsured farmers reduce debt persistently after the shock. The frost destroys their capital and lowers the value that they can pledge, so they borrow less even though the marginal return to replacing capital is high. Insured farmers instead expand debt to finance replanting.

Figure 2: Model-Implied Debt Responses: Insured and Uninsured Farmers

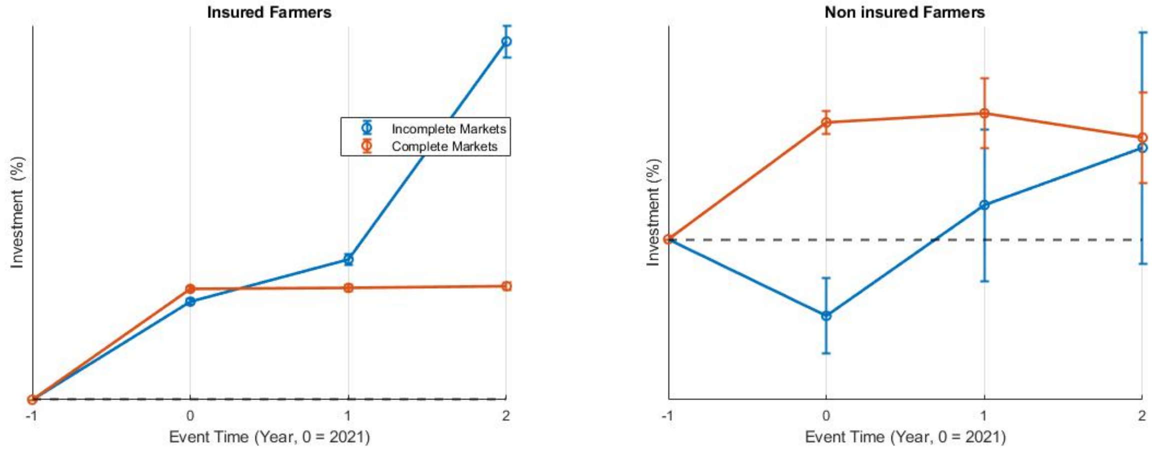


(a) Insured Farmers (b) Uninsured Farmers

Notes: The event-study plots are generated from model-simulated data using equation (20). We draw 1,000 farmers in each initial insurance group and randomly assign half of each group to receive the capital shock. The plots show cumulative effects on total outstanding debt relative to the period before the shock.

The investment responses follow the same pattern. Figure 3 shows that insured farmers increase investment to replace damaged capital. Uninsured farmers reduce investment and recover slowly. This is the opposite of the complete-market response, under which the farmers who lose the most capital should invest the most because they have the highest marginal product of capital.

Figure 3: Model-Implied Investment Responses: Insured and Uninsured Farmers

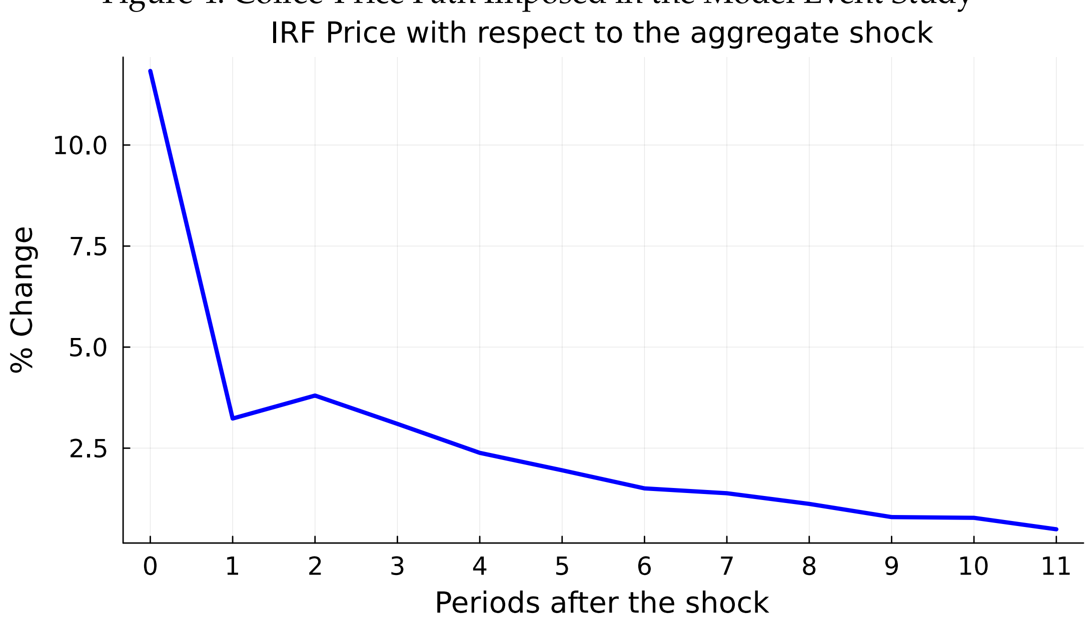


(a) Insured Farmers

(b) Uninsured Farmers

Notes: The event-study plots are generated from the same simulated panel used for Figure 2. The plots show cumulative effects on investment for insured and uninsured farmers relative to the period before the shock.

Figure 4: Coffee-Price Path Imposed in the Model Event Study



Notes: The plot shows the coffee-price path imposed in the solved fixed-price CPR model event study. It is not an equilibrium response. The general equilibrium model determines the price from aggregate output and inverse demand.

The coffee-price path in the fixed-price CPR event study affects both output revenue and pledgeable harvest value. It offsets part of the revenue and net-worth loss for farmers with surviving production and directly relaxes the CPR borrowing limit. The price can therefore widen the difference between affected farmers and farmers whose capital survives. Figure 4 shows the price path imposed in this exercise.

The model event studies give us the empirical predictions that we take to the data. After the shock, uninsured farmers reduce debt and investment, while insured farmers borrow and invest to replace damaged capital. We next estimate the same dynamic responses using variation in frost damage across nearby farms.

3 Setting and Data

Firms demand insurance due to revenue volatility and the desire to hedge against risk, which can directly arise from preferences or indirectly arise from the cost of default. While price insurance exists for commodities through derivatives markets, there are no available revenue insurance contracts in most sectors. This could be due to a combination of unraveling induced by adverse selection or moral hazard, credit constraints to pay the insurance premium upfront, and limited commitment by insurers to truthfully fulfill claims. In many countries, including the US and our setting in Brazil, the government offers insurance not only for farmers' revenues, but also for yields (quantities) and costs (input expenditures). Despite the widespread provision of subsidies, with insurance premiums an average of 30% below actuarially fair levels in Brazil, insurance take-up is as low as 20% for crops like coffee. While salience and search costs may be large enough frictions to result in low take-up among small "household" farmers, low take-up is common even among large "industrial" farmers in Brazil.

In addition, agriculture is a setting where liquidity shocks are particularly salient because farmers receive most of their income around harvest time. A shock that occurs between harvests can leave farmers in a difficult situation: current and future income decrease, but current expenditure increases in response to the shock. Even if the farmers have sufficient assets, they can face a cash flow shortage. This suggests that farmers' insurance take-up decisions are inherently linked to their credit access and borrowing constraints.

3.1 Data

There are three components to our main dataset: the rural credit registry, crop insurance, and other financial data. The Rural Credit Registry (SICOR) provides contract-level data for agricultural credit and the linked public crop insurance program. The contracts list the crop, the area, the expected production, the actual production, the insurance premium, the insured value, the coverage level, and the precise coordinates of the farm. We observe insurance claims that quantify the amount of the loss, the amount paid to farmers, and a description of the reason for the insurance payout. We supplement the credit and insurance data from SICOR with data from the Ministry of Agriculture (MAPA) on insurance contracts from the main private crop insurers. In MAPA, we observe similar variables to SICOR for premiums, claims, and farm characteristics. To measure non-insurance hedging, we use data on foreign exchange futures and derivatives contracts, which are cleared and registered through the Brazilian exchange B3, and reported to the Central Bank of Brazil (BCB). Coffee-specific contracts exist because coffee is an internationally traded commodity.

For firms' supply chain linkages and borrowing from banks, we merge transaction-level datasets of interfirm payments, invoices, and credit operations from the BCB. The BCB credit registry (SCR) contains all firms and individuals with total debt since June 2016 of over 200 BRL, equal to around 40 USD. Using the crosswalk between the BCB datasets and the administrative registry from the federal revenue service, we observe each counterparty's municipality and 7-digit CNAE code, which is similar in specificity to a 6-digit Harmonized System (HS) code. We classify specific CNAE codes as upstream of coffee farming, based both on the CNAE descriptions and payment shares, and split the sectors into investment purchases (e.g. seedlings) and material purchases (e.g. fertilizer). The SCR also contains information on the guarantee associated with each credit operation, which allows us to classify farmers according to the predominant guarantee on their borrowing before the frost, distinguishing harvest collateral such as CPR and *penhor* from real estate pledged through a mortgage or fiduciary transfer. See Appendix B for more details.

A key challenge in constructing the combined dataset of contracts at the farmer level was the creation of a crosswalk across insurance and credit contracts that small-scale farmers took under their personal identifier rather than their firm identifier. We believe that we are among the first to construct such a crosswalk for all farmers in any country.

For municipality-level statistics on weather and farm output that we only feature in Appendix A, we merge weather station data from Brazil's National Institute of Meteorology to agricultural survey data from the Municipal Agricultural Production (PAM) and Systematic Survey of Agricultural Production (LSPA) from the Brazilian Institute of Geography and Statistics (IBGE).

Table 3: Summary Statistics on the Frost Shock, Coffee Production, and Hedging

	Mean	25th Percentile	Median	75th Percentile	Number of Coffee Farmers
Farm Area (Ha)					
Insured	42.61	6.56	14.22	34.9	20,472
Uninsured	23.88	2.62	5.45	14.08	80,376
Revenue Share from Coffee (%)					
Insured	92%	97%	100%	100%	20,472
Uninsured	91%	100%	100%	100%	80,376
Frost Shock Magnitude (Damage Share)					
Insured	26%	-	-	-	20,472
Uninsured	27%	-	-	-	80,376
Insured Value (Thousand USD)					
Insured	122	25	49	125	20,472
Uninsured	-	-	-	-	80,376
Claim Payout (Share of Total Insured Value)					
Insured	30%	5%	20%	46%	20,472
Uninsured	-	-	-	-	80,376
FX Hedge (Share of Farmers)					
Insured	1.4%	-	-	-	20,472
Uninsured	0.06%	-	-	-	80,376
Outstanding Debt (Thousand USD)					
Insured	80	0	45	112	20,472
Uninsured	31	0	5.6	32	80,376
Default Rate (Share of Debt Over 90 Days Overdue)					
Insured	2.5%	-	-	-	20,472
Uninsured	6.2%	-	-	-	80,376
Interest Rate (per Annum)					
Insured	5.6%	1.1%	4.4%	6.5%	15,747
Uninsured	9.5%	2%	4.2%	5.5%	60,760
Outstanding Debt Maturity (Years)					
Insured	7.5	5.9	7.3	8.8	15,747
Uninsured	8.1	5.9	8.8	9.9	60,760
Annual Payment Outflows (Thousand USD)					
Insured	743	11	43	191	20,472
Uninsured	275	1.6	5.58	18.6	80,376
Agricultural Inputs Payments (Share of Total Payments)					
Insured	48%	20%	48%	75%	20,472
Uninsured	45%	11%	45%	78%	80,376

Notes: These summary statistics use data from the BCB and SICOR. The damage share of coffee plants is the insurer's assessment of the share of future coffee bean production that is lost due to damage from the frost. Insured value and claims are only available for insured farmers; the values are zero and undefined for uninsured farmers. The interest rates are low because the government heavily subsidizes agricultural lending in Brazil.

Table 3 shows summary statistics from our main dataset. Insured and uninsured farmers differ along most dimensions: uninsured farmers pay higher interest rates, default at higher rates, have smaller farms, and receive less revenue.[2] However, insured and unin-

[2]We define default the standard way in Brazil: debt that is unpaid at least 90 days after the due date.

sured farmers were similarly affected by the frost shock; there is little correlation between the frost shock magnitude and any systematic differences in insurance take-up across the country or within small geographical regions.

3.2 Frost Shock

We study the July 2021 frost in Brazil, which led to an unexpected decrease in the capital stock for affected coffee farmers. Three severe waves of frost affected the states of São Paulo, Minas Gerais, and Paraná between the end of June 2021 and the beginning of August 2021. The frost was multiple standard deviations outside the distribution of outcomes that farmers typically consider during planting season. Due to timing and geography, the frost primarily affected perennial crops like oranges and coffee, rather than annual crops like soybeans and maize. As a result, estimating the aggregate economic impact of the frost shock would be difficult from harvest data alone[3]. Measuring frost with weather data can be imprecise because small differences in geography and soil, in combination with small differences in temperature, can correspond to large differences in losses.[4]

We interpret the frost as a capital stock shock, rather than solely a productivity or cash flow shock, because coffee is a perennial crop. The frost killed coffee plants, whose value is included in the collateral that farmers post for loans. The timing of the frost, at the end of the harvest season, meant that the 2021 coffee harvest was not heavily disrupted. We define the shock at the farmer-crop level using insurance claims as a proportion of insured value. We believe that insurance claims are the best available granular metric of the shock, and better than the typical weather-based or region-level output-based metrics in the literature. Insurance claims are only paid after an agronomist visits the plot and validates the extent of the damages. We observe the cause of each claim, and we observe the precise coordinates of each insured plot.[5]

We define the frost shock metric FS at the farmer level by matching each farmer i who grew coffee to nearby insured coffee farmers $j \in \mathcal{N}_i$ who received an insurance claim only for frost in 2021. We observe farm coordinates x_i and, for each insured neighbor j, the ratio I_j of the frost insurance claim to insured value. We define the frost shock for farmer

[3]Indeed, the challenge of estimating the effect of frost without detailed microdata at the coffee farm level has been recognized for a long time. See, for example, Stevens (1955) for a historical discussion.

[4]See Appendix A for more details about the measurement of the frost shock and the geographical distribution of crops in Brazil.

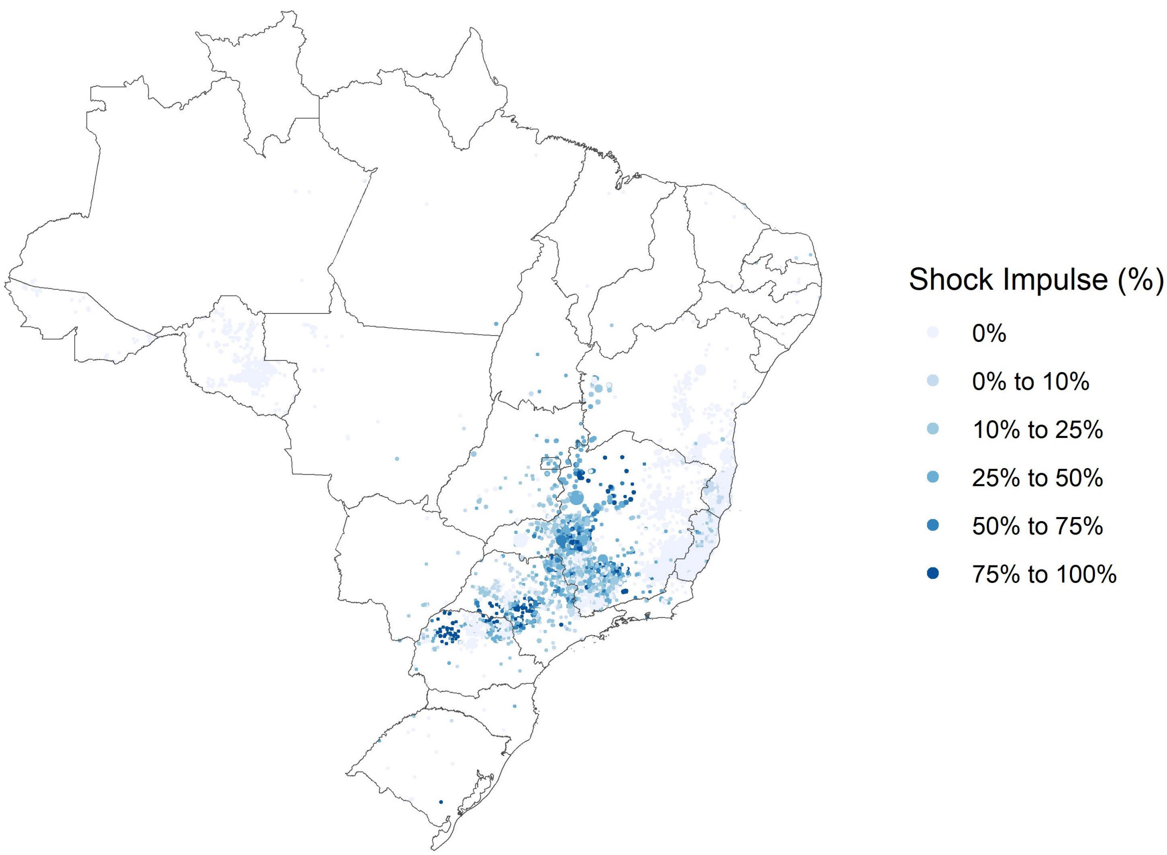
[5]The fact that insurance can be contingent on a frost shock is unsurprising, as frost events, though rare, tend to have significant impacts on coffee production in Brazil. There were at least 17 frost shock events in the 20th century.

i as

$$FS_i = \frac{\sum_{j \in \mathcal{N}_i} w_{ij}(x_i) I_j}{\sum_{j \in \mathcal{N}_i} w_{ij}(x_i)}, \quad w_{ij}(x_i) = \frac{1}{d(x_i, x_j)^\beta}, \tag{21}$$

where I_j is the insurance claim normalized as a share of underlying value for insured farmer j, $d(x_i, x_j)$ is the distance between farms, and $\mathcal{N}_i$ is the set of k-nearest insured neighbors of farmer i. Note that (21) is well-defined for all farmers i, regardless of whether farmer i purchased insurance. Both β and k are chosen by cross-validation.

Figure 5: Frost Shock Impulse Based on Coffee Insurance Claims



Notes: The insurance data are from SICOR. Each dot represents a farm that grew coffee in 2021. The size of the dot scales in proportion to the size of the farm. The color of the dot represents the shock impulse, equal to the magnitude of the shock, as measured by the proportion of coffee plants that were damaged, where darker colors represent higher magnitudes. The coffee-growing regions that were unaffected by the frost shock, with shock impulse equal to 0%, were Rondônia, eastern Minas Gerais, Espírito Santo, and eastern Bahía; these regions can be seen clearly in Figure A1 in Appendix A.

Figure 5 shows the geographical distribution of the frost shock. Although some coffee-growing regions (e.g. northern Paraná through northwestern Minas Gerais) were affected more than others (e.g. eastern Minas Gerais through Bahia),[6] there was variation in the severity of the frost shock even within each municipality, an administrative division that is on average half the size of a county in the contiguous US.

4 Empirical Methodology and Results

The model event study motivates our empirical specification. In the model, we compare farmers who receive the capital shock with farmers in the same initial insurance group who do not. In the data, we replace the simulated shock indicator with the continuous farm-level measure of frost damage and make the same comparison separately for insured and uninsured farmers. We do not compare insured farmers directly with uninsured farmers because insurance status is not randomly assigned.

Our identifying assumption is that the granular impulse of the frost was as good as random, given the outcome-specific fixed effects stated below. Our justification is that the frost was not anticipated at the time of planting, the most severe since the advent of modern weather stations, and the weather forecast did not predict it until a week beforehand, after all planting and insurance decisions were sunk. Even then, forecasts did not capture the granular spatial variation in the realized shock.

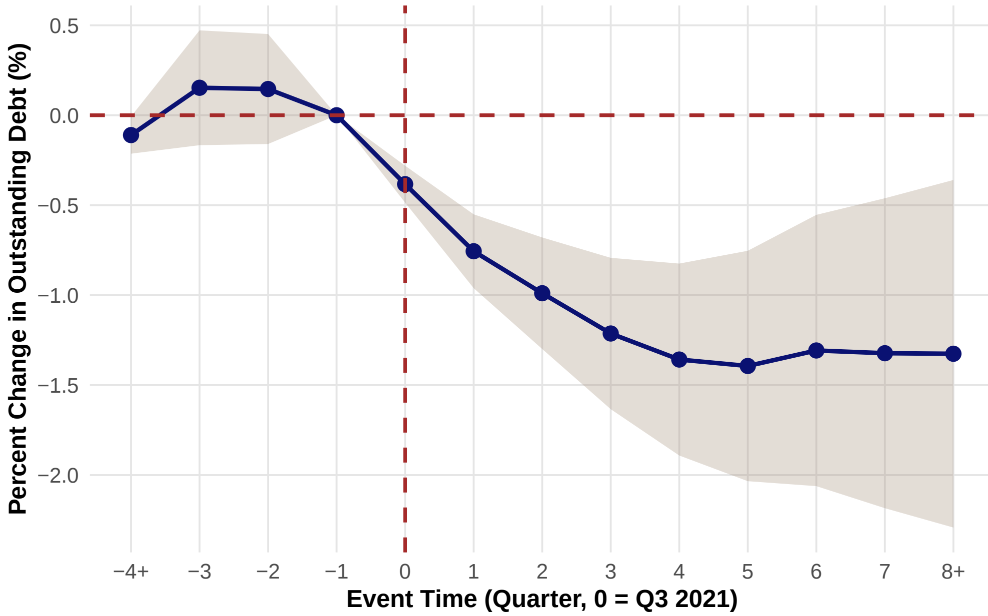
Our regressions encompass three sets of outcome variables. The credit regressions examine affected farmers' outcomes y_{ijt}, such as outstanding debt and maturity, with separate event study coefficients by whether the farmers had existing insurance contracts in place. Our fixed effects control for farm characteristics as well as other shocks that affected any combination of lender, location, and time. Let i be farmer, j be municipality, let g denote insured versus uninsured, and t be quarter. Let $s_{ij\tau}$ be the interaction of our preferred shock metric FS_i, defined in equation (21), with an indicator for whether the shock occurred in the given period: $s_{ij\tau} = \mathrm{FS}_i \mathbb{1}\{t = \tau\}$. Let Ins_i be an indicator of whether farmer i purchased insurance at the beginning of the growing season of the shock.

$$y_{ijt} = \sum_{\tau=2019Q1}^{2023Q4} \beta_\tau^C s_{ij\tau} \mathrm{Ins}_i + \sum_{\tau=2019Q1}^{2023Q4} \gamma_\tau^C s_{ij\tau} \mathrm{No\_Ins}_i + \alpha_i + \alpha_{jgt} + \epsilon_{ijt}. \tag{22}$$

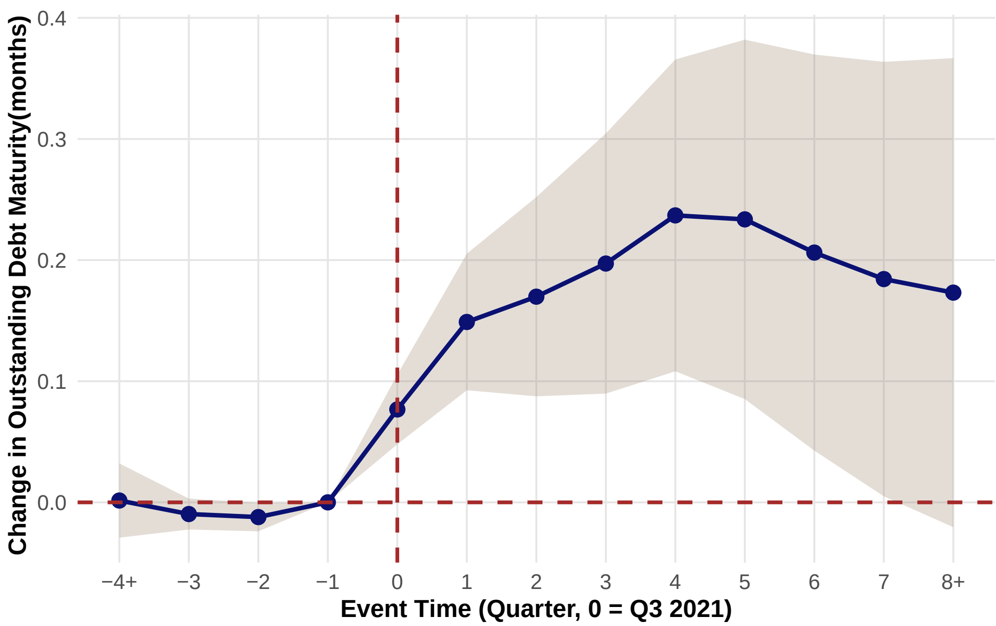
In other words, the β_τ^C coefficients in equation (22) compare shocked insured farmers to non-shocked insured farmers, and the γ_τ^C coefficients compare shocked uninsured farmers to non-shocked uninsured farmers. Our rationale is that conditional on the granular fixed effects, the magnitude of the frost shock was as good as random, and differences in outcomes are reflective of the differences in damages through farmers' reduced net worth.

[6] See Figure A1 in Appendix A for a map of coffee production in Brazil.

Figure 6: Credit Regression Results of $\{\gamma_\tau^C\}$ for Uninsured Farmers



(a) Outstanding debt decreased



(b) Debt maturity increased

Notes: These regressions use credit registry data from the BCB. The event study coefficients correspond to the cumulative effects across $\{\gamma_\tau^C\}$ in equation (22), comparing credit outcomes for shocked uninsured farmers to non-shocked uninsured farmers. The interpretation of the shock magnitude $s_{ij\tau}$ is one percentage point increase in coffee plant damage. The outcome variables y_{ijt} are outstanding debt balance in panel (a) and outstanding debt maturity in panel (b).

Figure 6 shows the event study coefficients for the $\{\gamma_\tau^C\}$ coefficients, across event time τ

on the horizontal axis, for the outcome y_{ijt} outstanding debt balance in Figure 6a and the outcome y_{ijt} debt maturity in Figure 6b.[7] The takeaway from Figure 6 is that each additional percentage point of coffee plant damage for uninsured farmers, comparing against uninsured farmers with zero damage, corresponds to a 0.4% immediate reduction and 1.4% long-run reduction in outstanding debt balance. Our interpretation is that farmers' net worth decreased, so their borrowing constraints tightened and reduced their debt load despite having higher marginal returns to capital. The takeaway from Figure 6b is that each additional percentage point of coffee plant damage for uninsured farmers, comparing against uninsured farmers with zero damage, corresponds to a 0.08 month immediate increase and 0.2 month long-run increase in outstanding debt maturity. Our interpretation is that farmers renegotiated debt and received forbearance on their loans.

By comparison, we show in Figure A4 in Appendix A that the results for $\{\beta^C_\tau\}$ in regression (22) are zero across all event time, as we expect; insurance cushions farmers from the financial impacts of the shock. In addition, we find no significant effects on other credit outcomes: the debt interest rate, the default rate, and the composition of farmers' debt.

The payment regressions examine how farmers' purchases changed, specifically focusing on purchases from suppliers in sectors that are consistent with investment in regrowing crops or in the quality of the land.[8]

$$\begin{aligned} y_{ijst} = &\sum_{\tau=2019Q1}^{2023Q4} \beta^P_\tau s_{ij\tau} \mathrm{Ins}_i \mathrm{Upstream}_s + \sum_{\tau=2019Q1}^{2023Q4} \gamma^P_\tau s_{ij\tau} \mathrm{No\_Ins}_i \mathrm{Upstream}_s \\ &+ \alpha_{is} + \alpha_{jmt} + \epsilon_{ijst}, \end{aligned} \tag{23}$$

where subscript i is farmer, j is municipality, s is the supplier, t and τ are at the quarterly level, and m is the supplier's 7-digit CNAE code. The fixed effects control for farmer-supplier linkages as well as other shocks that affect any combination of municipality, time, and supplier sector. The coefficients identify differential responses to frost damage within these fixed effects, separately by initial insurance status. The outcome y_{ijst} is log payment *outflows* to suppliers of agricultural inputs (e.g. seeds and fertilizers) and similar agricultural services that are consistent with investments to increase future farm output. Similarly to the credit regressions, we do not compare insured to uninsured farmers because insurance status is not randomly assigned.

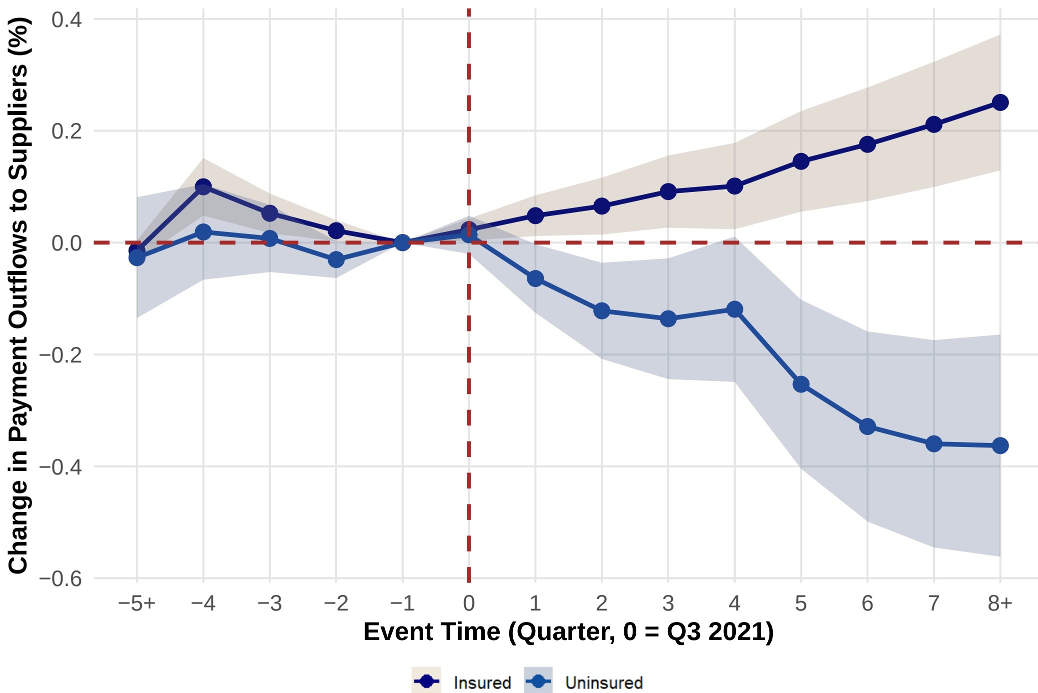
Figure 7 shows the event study coefficients for the $\{\gamma^P_\tau\}$ and $\{\beta^P_\tau\}$ coefficients from equation (23), across event time τ on the horizontal axis. For each percentage point of additional damage, affected non-insured farmers decrease investment over time compared to

[7] See Figure A4 in Appendix A for the results for the $\{\beta^C_\tau\}$ coefficients.

[8] We do not use payments to proxy for farmers' sales because many of the farmers' sales are made through advance purchase (CPR) contracts that are not directly affected by the shock.

unaffected non-insured farmers, with a long-run decrease of 0.38 percent. Our interpretation is that marginal investments are likely to be financed, and farmers' borrowing constraints tighten as net worth decreases due to the shock. By comparison, affected insured farmers increase investment over time compared to unaffected insured farmers, with a long-run increase of 0.24 percent per 1 percentage point of damage. We believe that this is due to the liquidity injection from the timing of the insurance payout, which is typically as soon as an insurance adjuster can reach the farm, versus typical farm revenues that occur months later.

Figure 7: Payment Outflow (Investment) Regression Results of $\{\gamma^P_\tau\}$ and $\{\beta^P_\tau\}$



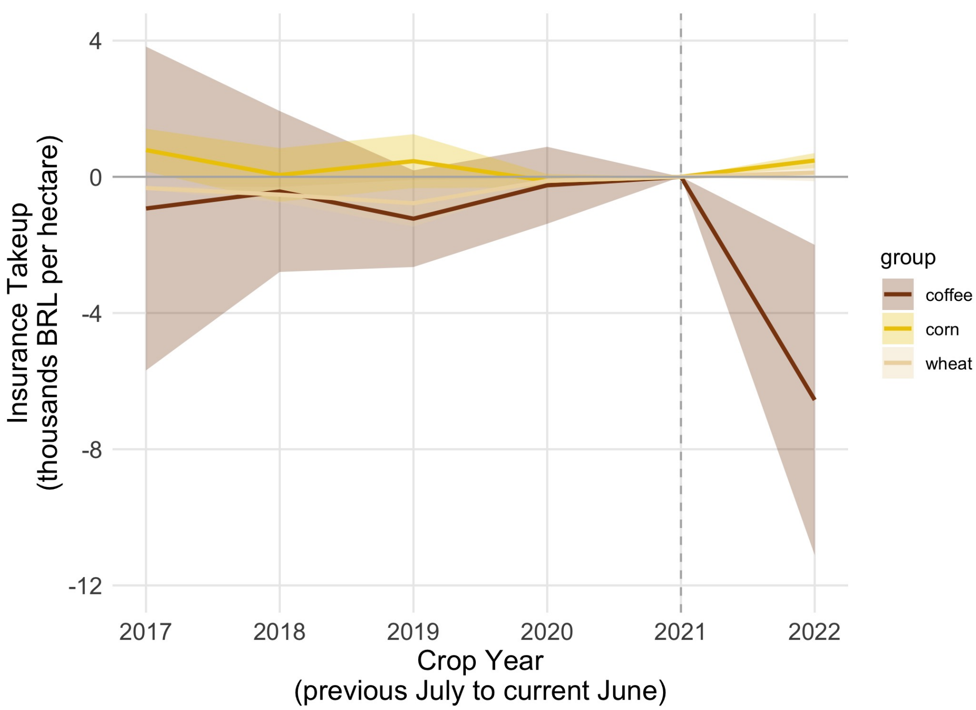
Notes: These regressions use payment data from the BCB. The event study coefficients correspond to the cumulative effects across $\{\gamma^P_\tau\}$ and $\{\beta^P_\tau\}$ in equation (23). Coefficients $\{\gamma^P_\tau\}$ compare shocked uninsured farmers to non-shocked uninsured farmers, where shocked uninsured farmers decreased investment more, particularly in the next growing season 4-8 quarters later. Coefficients $\{\beta^P_\tau\}$ compare shocked insured farmers to non-shocked insured farmers, where shocked insured farmers invested more. The interpretation of the shock magnitude $s_{ij\tau}$ is one percentage point increase in coffee plant damage.

The insurance regression in equation (24) examines how farmers' subsequent uptake of insurance and other hedging changed in response to the shock:

$$y_{ijct} = \sum_{\tau=2017}^{2022} \beta^I_{c\tau} s_{ijc\tau} \mathrm{Ins}_{ic} + \alpha_i + \alpha_{jct} + \epsilon_{ijct}, \tag{24}$$

where i is farmer, c is crop, j is municipality, t and τ are at the annual (growing season) level, and Ins_{ic} is an indicator for whether farmer i's plot of crop c was insured at the time of the frost shock. The fixed effects control for farm characteristics as well as any other shocks that affect any combination of municipality, crop, and time. The coefficients $\beta^{I}_{c\tau}$ compare more and less affected farmers within the group insured at the time of the frost. This specification does not estimate a separate subsequent take-up path for farmers who were initially uninsured. Figure 8 shows that, among farmers insured at the time of the frost, affected farmers subsequently took up less insurance than unaffected farmers for coffee, which is a perennial crop where the shock had lasting effects on net worth. However, subsequent insurance take-up for the annual crops corn and wheat were unchanged, since farmers plant new seeds every growing season, even though insured losses for corn and wheat were substantial.[9] In the model, we rationalize the subsequent decrease in coffee insurance take-up among farmers who experienced the shock through the lens of underinsurance as an endogenous consequence of net worth shocks in the presence of financial constraints.

Figure 8: Insurance Take-up Comparing Frost-Affected Farmers to Unaffected



Notes: The regressions are run on insurance data from SICOR and correspond to equation (24) for each of the three main crops c affected by the frost shock: corn, coffee, and wheat. The coefficients compare more and less affected farmers among those insured at the time of the frost. Only coffee has a significant insurance response because coffee is a perennial crop, for which the frost shock reduced affected farmers' net worth, while corn and wheat are annual crops.

[9]Note that soybeans, the primary crop in Brazil, were unaffected by the frost shock because of the timing of the soybean growing season.

In Appendix A, we show additional results that complement the main results: we find no significant differences in default rates in Figure A6, interest rates in Figure A5, or insurance premiums in Figure A7 between affected and unaffected farmers, in both the insured and uninsured groups. We interpret these findings as consistent with a setting where the financial market anticipates farmers' ability to repay debt in the event of shocks and allows for some form of state-contingent repayment. This may occur through mechanisms such as mandatory insurance or automatic debt renegotiation, as observed in certain rural credit contracts. However, this does not imply complete risk-sharing. As the model shows, limited commitment constrains both insurance and credit uptake.

In conclusion, the frost shock led to a reduction in outstanding debt among uninsured farmers, with an increase in debt maturity through renegotiation and forbearance. After the shock, affected uninsured farmers borrowed from subsidized credit lines to smooth consumption, but did not increase expenditure on investment goods, unlike affected insured farmers. We also find that affected uninsured farmers reduced labor demand, both on the intensive and extensive margin, unlike affected insured farmers. Total value insured decreased for affected farmers but increased for nearby unaffected ones. The credit channel alone was insufficient to compensate for the lack of insurance, even though emergency credit lines were extended in the period. See Appendix A for more details.

These results follow the model event-study predictions. Uninsured farmers reduce debt and investment even though the shock increases their marginal return to replacing capital, while insured farmers invest to rebuild. The debt-maturity response also shows that state contingency can arise through renegotiation even when observed interest rates and default do not change.

5 Policy Implications

In this section, we use the CPR and land implementation of the model, which sets $\lambda_K = 0$.

5.1 Emergency Credit Lines

We examine the effectiveness of emergency credit lines within the model's framework. Following the frost shock, the Central Bank of Brazil allocated 1.3 billion BRL (approximately 250 million USD) to be disbursed as credit lines to coffee farmers impacted by the event. Financial intermediaries, such as banks and credit cooperatives, were responsible for disbursing the funds. However, only 49% of the total amount was actually lent, and according to credit registry microdata, a disproportionately high share went to farmers already holding some form of quantity insurance: 40% of all lending went to the 20% of farmers who had purchased insurance.[10] Given the extent of the frost shock, why did the emergency credit not reach those who needed it most?

The model highlights a key channel: an emergency credit line does not increase borrowing capacity when it is subject to the same collateral constraint as ordinary rural credit. Let L_i denote the emergency loan disbursed today, let $R_L L_i$ be its promised repayment next period, and let $\overline{L}_i$ be the loan limit. The current budget becomes

$$c_i = W_i + \mathbb{E}\left[\frac{b_i(S', z')}{R}\right] + L_i - k_i. \tag{25}$$

If ordinary and emergency credit use the same collateral, then

$$b_i(S', z') + R_L L_i \leqslant \lambda p(S') A_i \theta_i(S', z')^\alpha k_i^\alpha + \chi_i q_i(S', z'), \qquad 0 \leqslant L_i \leqslant \overline{L}_i, \quad \forall (S', z'). \tag{26}$$

Increasing the announced limit $\overline{L}_i$ does not relax a collateral constraint that already binds. As a result, the additional funds might not be disbursed or flow toward insured and otherwise less constrained farmers. Emergency credit cannot replace insurance when both depend on the same pledgeable collateral.

A government guarantee changes this result. Suppose that the guarantee covers fraction ϕ_G of the emergency loan's promised repayment. The borrowing constraint becomes

$$\begin{aligned} b_i(S', z') + (1 - \phi_G) R_L L_i \leqslant \lambda p(S') A_i \theta_i(S', z')^\alpha k_i^\alpha \\ + \chi_i q_i(S', z'), \\ 0 \leqslant L_i \leqslant \overline{L}_i, \qquad \phi_G \in [0, 1]. \end{aligned} \tag{27}$$

Only the non-guaranteed share $(1 - \phi_G) R_L L_i$ uses the farmer's collateral capacity. In the model, increasing the effective guarantee ϕ_G, rather than the announced volume alone, relaxes the farmer's collateral constraint. Other policies can relax the two collateral constraints directly. Better registration, verification, or monitoring of CPR proceeds raises harvest pledgeability λ, while better land titles, registries, or enforcement raise land pledgeability χ.

The low disbursement rate is consistent with the model because emergency credit is least likely to reach constrained farmers when it has the same collateral requirements as ordinary rural credit.

These results are related to the experience of the Paycheck Protection Program in the United States. Joaquim and Wang (2022) show that emergency lending through financial intermediaries did not always reach the firms most affected by the COVID crisis. In

[10] For the total amount allocated, see Resolution of the National Monetary Council No. 4,954 of October 21, 2021. For the total amount disbursed, see a report from the Ministry of Agriculture.

both settings, the announced availability of funds can differ from the ability of constrained firms to use them. Corporate emergency credit following the floods in Rio Grande do Sul provides another setting in which the model's distinction between announced liquidity and usable collateral capacity can be examined. We do not use that episode as evidence for the mechanism in this paper.

5.2 Increasing the Pledgeability of Collateral

The distinction between liquidity and pledgeability also clarifies the role of recent innovations in agricultural credit. Registration of CPR and CPR-F contracts, together with satellite or drone verification of crop conditions and harvest proceeds, can increase the fraction of future revenue that lenders can verify and recover. In the model, this is an increase in λ. Monitoring alone is not sufficient. The proceeds must be identifiable, diversion to unregistered buyers must be limited, and the lender must have an enforceable claim.

Better land titles and registries instead increase χ by making a larger share of land value available as collateral. Fractionalization or tokenization can have the same effect only if the registry prevents multiple pledges and establishes the priority of each lender's claim. It may also allow a farmer to pledge unused land value to another lender, although the resulting effect on lender competition is outside the model because the interest rate is exogenous.

As an illustrative policy extension, a monitored CPR payment account can increase effective pledgeability when registered crop-sale proceeds are verifiable and seizable. Let E_{it} denote the balance in the account and let m_{it} denote monitoring quality. The borrowing constraint becomes

$$b_i(S', z') \leqslant \lambda p(S') A_i \theta_i(S', z')^{\alpha} k_i^{\alpha} + \chi_i q_i(S', z') + \lambda_E(m_{it}) E_{it}, \qquad \lambda_E'(m) > 0. \tag{28}$$

If the account requires the farmer to hold funds in escrow, it also removes working capital. The benefit from a more pledgeable dollar must therefore be compared with the farmer's shadow value of internal liquidity and any benefit from coordinated renegotiation. The policy is most useful when it directs verified proceeds toward farmers whose collateral constraint is valuable to relax, without removing operating cash from farmers with the greatest need for internal liquidity. We therefore treat the increase in pledgeability as the policy object rather than assuming that monitoring or escrow is beneficial by itself.

5.3 Climate Stress Testing and the Incidence of Risk

The model also changes how we interpret climate stress tests. These exercises normally ask how much climate risk is borne by banks and insurers. In our setting, low financial-sector exposure need not mean that the underlying risk is small. When collateral constraints prevent farmers from borrowing or purchasing insurance, the risk remains with the farmers. They respond by reducing investment, which delays the recovery of coffee production. Under the maintained inelastic-demand benchmark, the reduction in aggregate supply raises the coffee price and passes part of the loss to consumers. The price increase also raises the value of surviving harvest and grove collateral, so the final effect on borrowing capacity depends on which farmers retain productive capital after the shock.

We define a climate stress scenario by changing the severity and frequency of the capital shock. Let $\mathcal{S}_{\text{stress}} \subset \mathcal{S}$ denote the aggregate stress states. A more severe stress scenario has

$$\theta_f^{\text{stress}} = (1 - \delta_{\text{stress}})\theta_n < \theta_f, \qquad \delta_{\text{stress}} \in (0, 1), \tag{29}$$

while a more frequent scenario replaces $\pi(S' \mid S)$ with a transition matrix $\pi^{\text{stress}}(S' \mid S)$ that assigns more probability to states with high frost incidence. The frost observed in 2021 provides a benchmark for the physical shock, while the transition matrix determines how often farmers expect similar states to occur.

We propose a stress test that solves the equilibrium under the baseline and stressed processes and compares insurance coverage, debt, investment, the share of farmers whose collateral constraint binds, the distribution of net worth, aggregate output, and the coffee price. The comparisons will be made under CPR collateral, fixed land collateral, and grove-inclusive land collateral. We will also run the same event-study regression on each simulated panel. This will show whether a change in climate risk affects only the initial physical loss or also changes the persistence and distribution of the response through insurance, collateral, and the coffee price.

The baseline model assumes that financial intermediaries have deep pockets. To study a contraction in insurance supply during an aggregate climate event, we also consider an extension in which primary insurers must cede a fixed fraction ρ of aggregate exposure to reinsurers whose stress-state capacity is $\bar{K}(S')$. We assume that primary insurers cannot reduce this required cession share when the constraint binds:

$$\int_0^1 \rho h_i(S', z_i') \, di \leqslant \bar{K}(S'), \qquad S' \in \mathcal{S}_{\text{stress}}. \tag{30}$$

When this constraint binds, farmers cannot purchase all of the state-contingent coverage they would otherwise choose. Because insurance supports borrowing in the model, the coverage limit can reduce debt and investment and increase the real-sector exposure to the shock.

This is the other side of climate stress testing. A stress test that reports only the direct losses of financial institutions can miss the losses borne by farmers, suppliers, and consumers when credit and insurance are rationed.

6 Conclusion

Limited collateral pledgeability provides a common explanation for three features of climate adaptation that are usually studied separately: producers under-insure before a disaster, post-disaster credit fails to reach the producers who were damaged most, and physical shocks propagate through supply chains. Insurance and production compete for the same borrowing capacity, so under-insurance is a constrained-optimal choice rather than a behavioral mistake. After a shock, uninsured farmers can have the highest marginal return to replacing capital but the least ability to finance it.

We extend the model to distinguish future harvest revenue pledged through a CPR from land collateral. Bare land can cushion a frost shock, whereas the part of land value that reflects standing coffee trees remains exposed to physical destruction. The same frost that destroys the asset raises the price of what survives, so the price increase affects both current revenue and the value of price-sensitive collateral. The general equilibrium effects have both a budget channel and a collateral channel, and an aggregate shock widens the gap between damaged and undamaged farmers rather than only shifting the level of borrowing.

The empirical results follow the model predictions. Affected uninsured coffee farmers reduce borrowing and investment and extend debt maturity. Insured farmers instead increase investment to replace damaged capital. Among farmers insured at the time of the frost, those more affected subsequently purchase less insurance than nearby less affected farmers. We find no corresponding increases in interest rates, default, or insurance premiums. The reduction in purchases from upstream sectors shows how under-insurance propagates a physical climate shock through the supply chain. The credit channel alone was insufficient to compensate for the lack of insurance.

These findings change what climate policy should target. Increasing the supply of intermediated credit is not equivalent to increasing the borrowing capacity of the farmers most affected by the shock, which is why less than half of Brazil's emergency credit line was disbursed and why most of the lending reached farmers who were already insured. Public guarantees and measures that raise the pledgeability of harvest proceeds and land act on the constraint that binds. The same point applies to climate stress testing. Low exposure of financial institutions can reflect an inability of farmers to transfer risk rather than resilience, leaving more of the loss with farmers, suppliers, and consumers through lower investment, lower output, and higher coffee prices.

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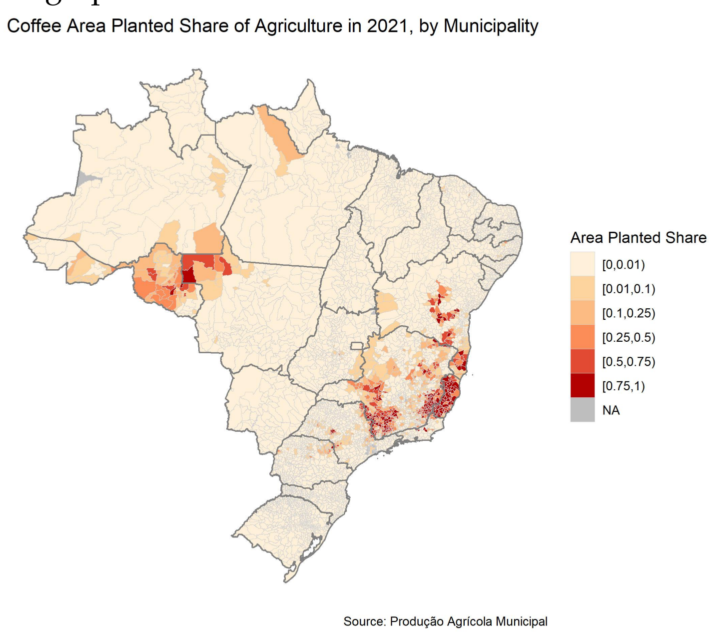
A Empirical Appendix

The purpose of this section is to provide additional context on coffee farming in Brazil and on the frost shock. Coffee is a perennial crop whose coffee beans are harvested from coffee cherries that typically grow on the plant each year. Coffee plants can live for many decades, and typically do not begin producing cherries until the third growing season. The growing season of coffee in south central Brazil begins in September to November, with harvest in June through August. Coffee plants are sensitive to cold temperatures, with low frost tolerance. The primary coffee growing regions in Brazil, depicted in red in Figure A1, are in regions where nighttime winter temperatures rarely drop below 10 Celsius.

Figure A1: Geographical Distribution of Coffee Production in Brazil in 2021



Notes: This figure uses data from the Municipal Agricultural Production (PAM) survey to plot the municipality-level share of agricultural land in 2021 where coffee was planted. In Brazil, the municipality is the administrative division that is roughly analogous to the county in the US.

However, during the frost shock, nighttime temperatures were lower than climatic averages by more than 10 degrees Celsius for multiple periods of consecutive days. The first frost did not appear in weather forecasts until 3 days before the frost on June 28, 2021, and anecdotes from news articles suggest that farmers had limited options to protect their trees, given the scale of coffee farming and the fragility of coffee plants. To assign the distribution of weather variables to each municipality, we follow the methodology of Deschênes and Greenstone (2007) and subsequent papers in environmental economics.

Figure A2: Locations of Weather Stations in Brazil in July 2021

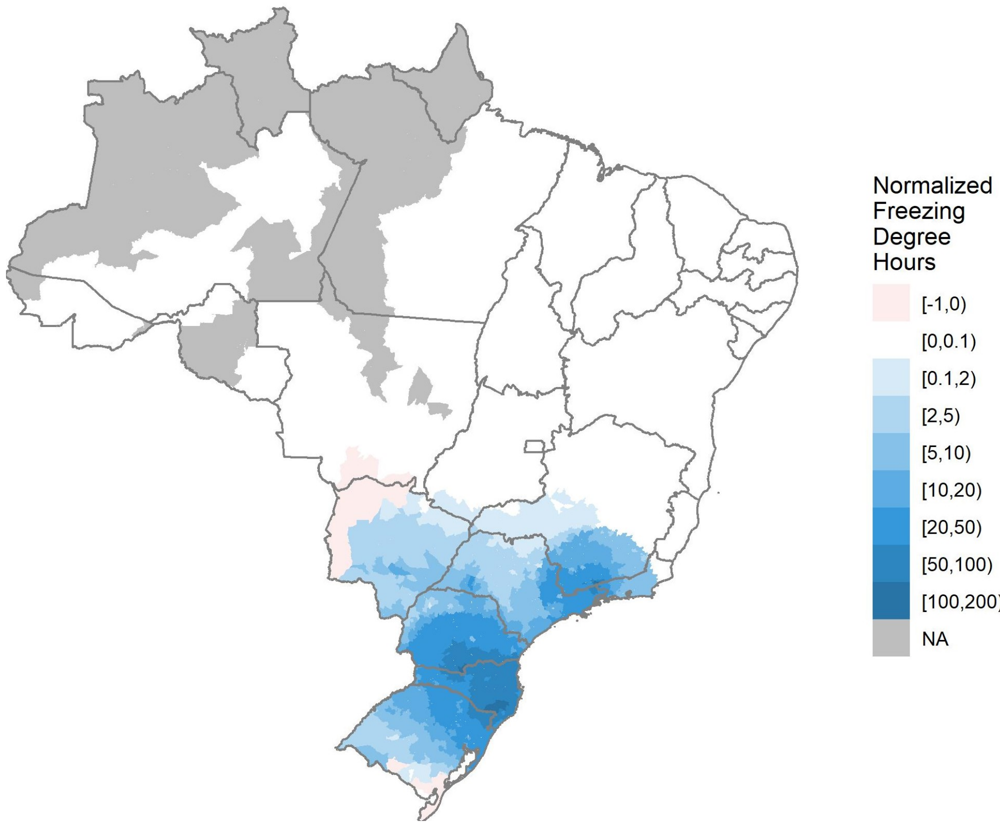
Notes: This figure shows the locations of weather stations from Brazil's National Institute of Meterology in yellow, as well as municipality centroids in black.

Using station-by-hour data from weather stations monitored by Brazil's National Institute of Meterology, whose distribution is shown in Figure A2, we count the number of hours over the frost shock period of June 28 through August 2, 2021 in each temperature bin of width one degree Celsius. Then, we use inverse distance weights $\omega_{s,i}$ between municipality i centroids and weather stations s with a 100 kilometer radius to define the municipality-level count of hours in each temperature bin. We define freezing-degree-hours (FDH) similar to growing-degree-days:

$$\mathrm{FDH}_i = \sum_{s \text{ near } i} \omega_{s,i} \sum_{\text{hour } h} \max\{0, -T_{s,h}\}. \tag{31}$$

where $T_{s,h}$ is mean hourly temperature at weather station s. Figure A3 shows the distribution of FDH_i across municipalities i with sufficient weather data. However, the sensitivity of coffee plants to cold weather is highly non-linear, which a quasi-linear variable like FDH cannot accurately capture. Coffee plants begin to experience defoliation and frost burn at temperatures as high as 5 Celsius, which a much wider swath of central Brazil experienced during the frost shock. The extent of damage varies with wind, and even the nutrient composition of the soil, in addition to temperature. As a result, we believe that Figure 5 captures the impulse of the frost shock more accurately than any weather-based metric, like Figure A3, that follows the standard methods from the environmental economics literature or the climate finance literature.

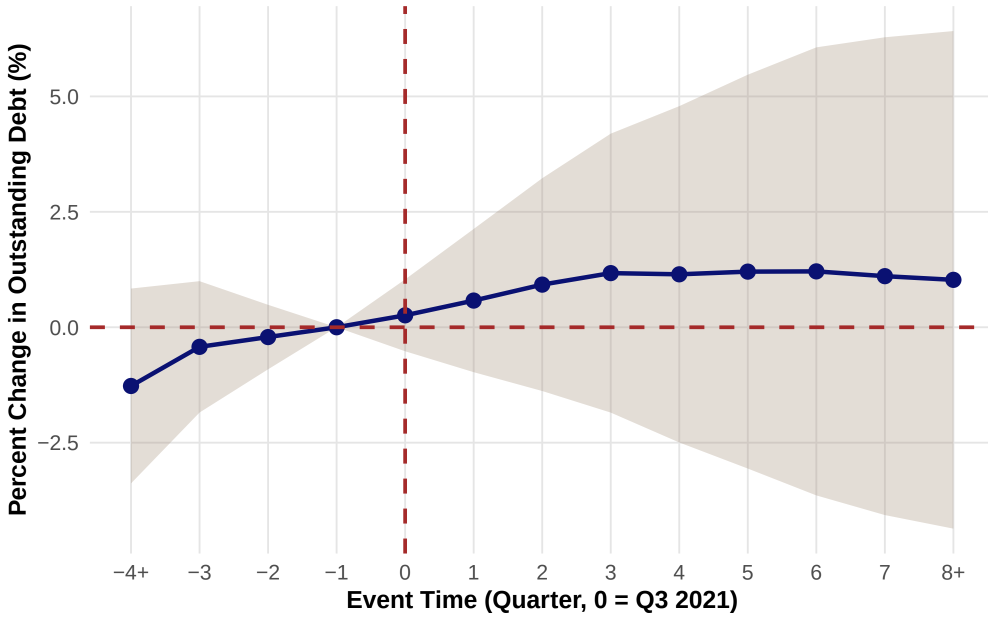
Figure A3: Municipality-Level Freezing-Degree-Hours During the Frost Shock Period



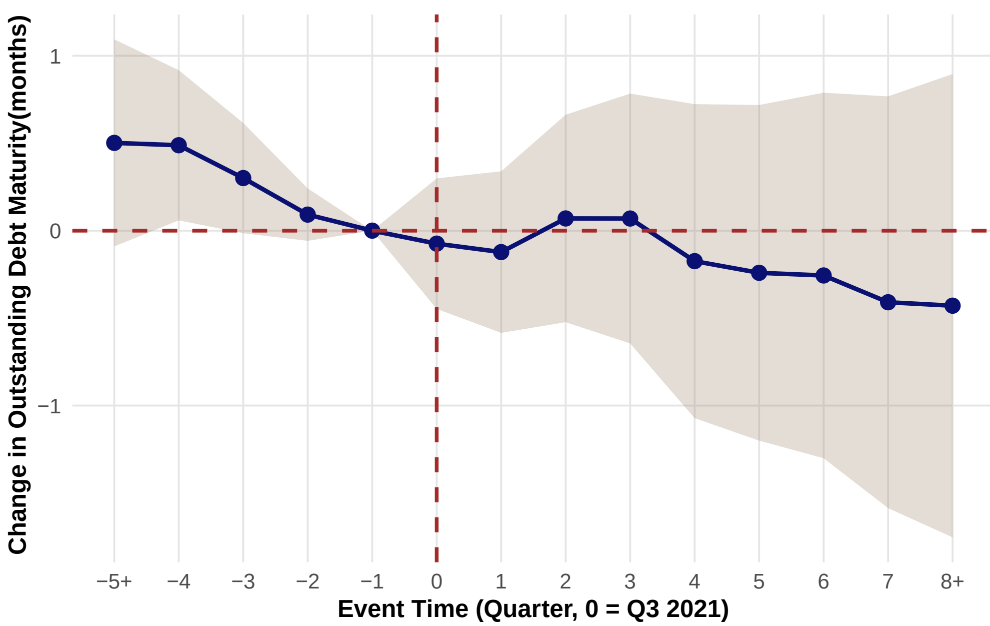
Notes: This figure shows the geographical distribution of freezing-degree-hours (FDH), normalized by the 2011-2020 average, in each municipality. We compute FDH following equation (31) using weather station data from Brazil's National Institute of Meterology.

A.1 Additional Event Studies Results

Figure A4: Credit Regression Results of $\{\beta^C_\tau\}$ for Insured Farmers



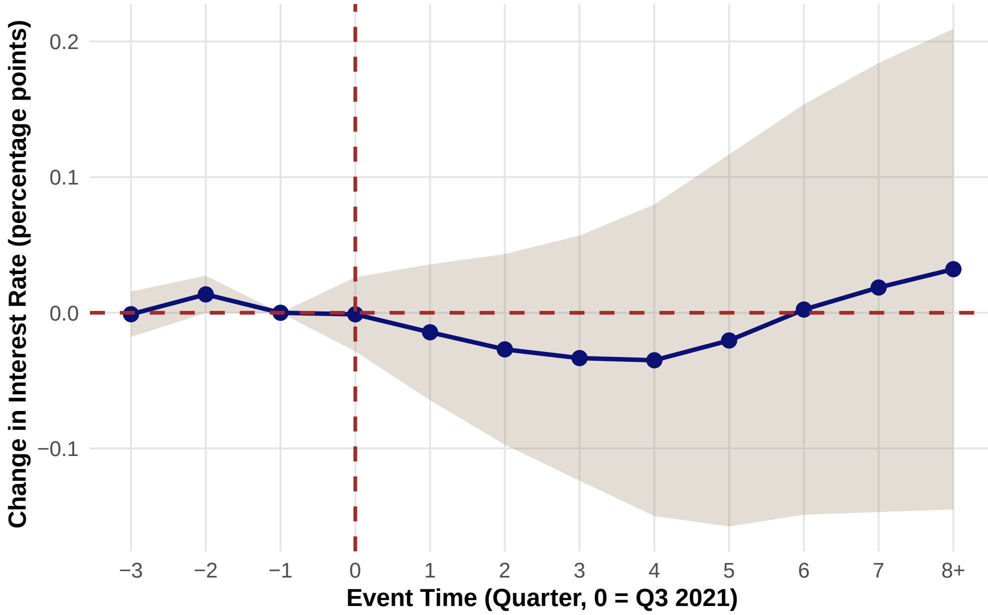
(a) No statistically significant effects on outstanding debt



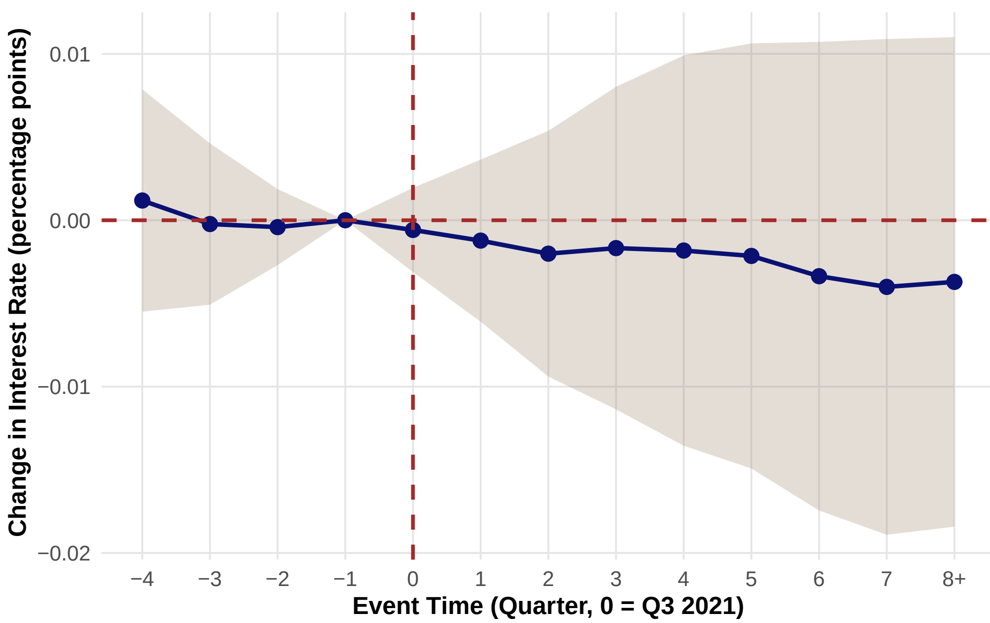
(b) No statistically significant effects on debt maturity

Notes: These regressions use credit registry data from the BCB. The event study coefficients correspond to the cumulative effects across $\{\beta^C_\tau\}$ in equation (22), comparing credit outcomes for shocked insured farmers to non-shocked insured farmers. The interpretation of the shock magnitude $s_{ij\tau}$ is one percentage point increase in coffee plant damage. The outcome variables y_{ijt} are outstanding debt balance in panel (a) and outstanding debt maturity in panel (b).

Figure A5: Interest Rate Regression Results for Uninsured Farmers and Insured Farmers



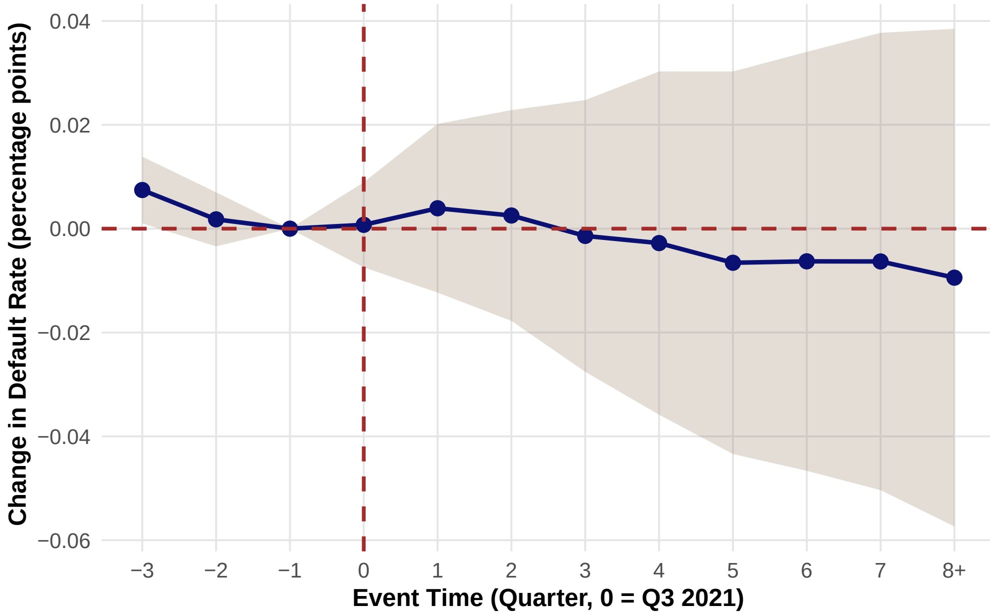
(a) No statistically significant effects on uninsured farmers' debt interest rate.



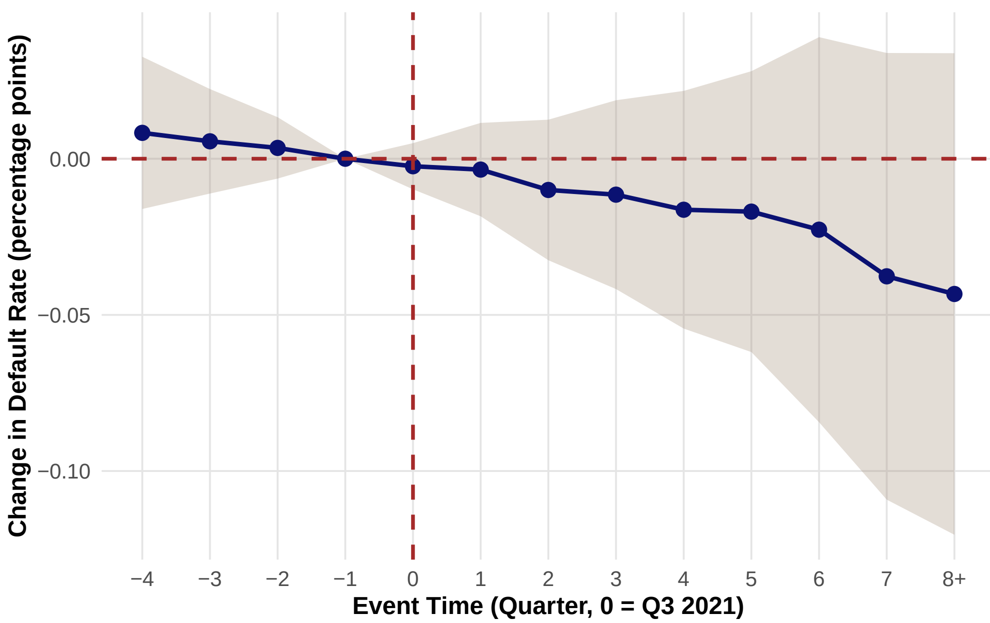
(b) No statistically significant effects on insured farmers' debt interest rate.

Notes: These regressions use credit registry data from the BCB. The event study coefficients represent the cumulative effects comparing the interest rates (on outstanding debt) for shocked farmers to non-shocked farmers. The shock magnitude corresponds to a one percentage point increase in coffee plant damage. The outcome variable, y_{ijt}, is the outstanding volume-weighted average interest rate on all farmers' credit operations. Plot (a) shows the results for uninsured farmers, while plot (b) shows the results for insured farmers.

Figure A6: Default Rate Results for Uninsured Farmers and Insured Farmers



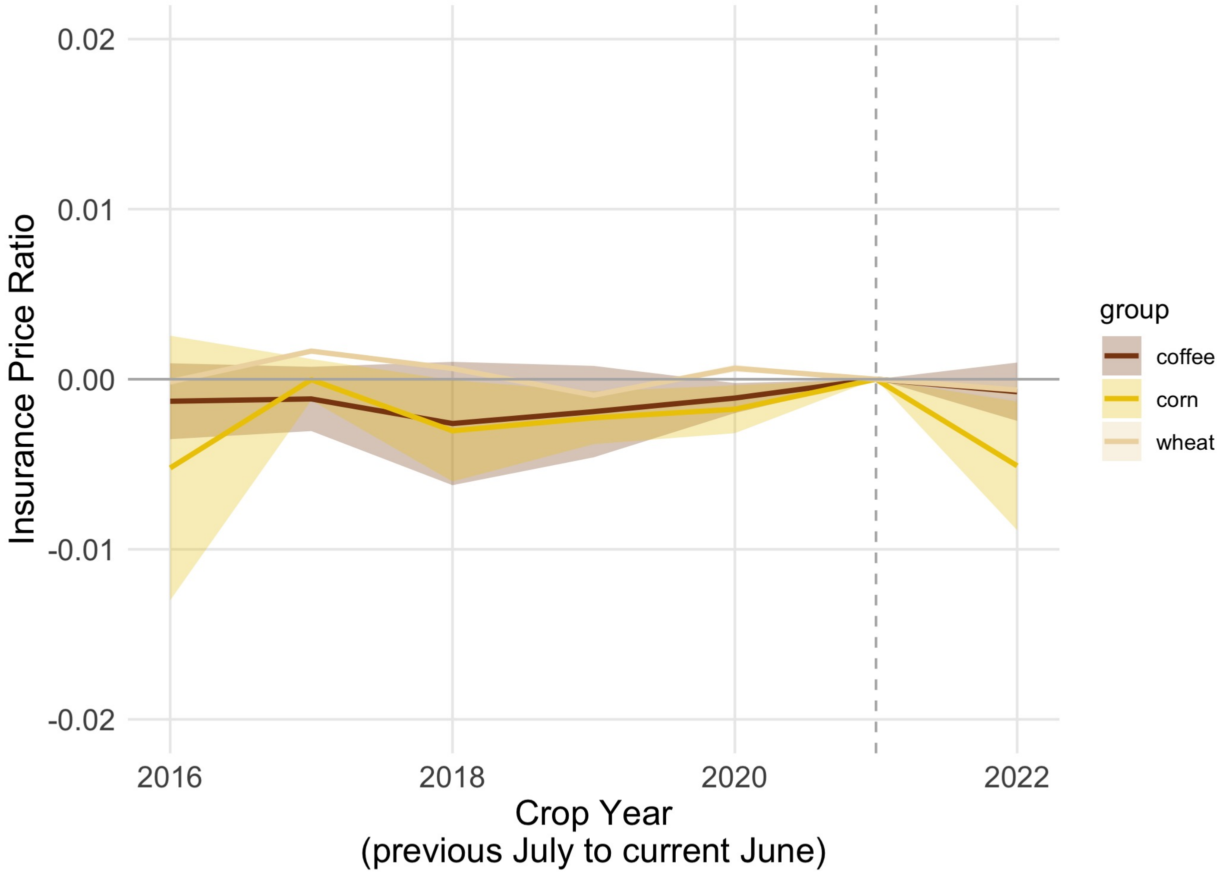
(a) No statistically significant effects on uninsured farmers' default rate.



(b) No statistically significant effects on insured farmers' default rate.

Notes: These regressions use credit registry data from the BCB. The event study coefficients represent the cumulative effects comparing the default rates (on outstanding debt) of shocked farmers to non-shocked farmers. The shock magnitude corresponds to a one percentage point increase in coffee plant damage. The outcome variable, y_{ijt}, is the default rate, defined as the total volume in default (more than 90 days late payments) divided by the total outstanding debt. Plot (a) shows the results for uninsured farmers, while plot (b) shows the results for insured farmers.

Figure A7: Insurance Premium of Frost-Affected Farmers to Unaffected



Notes: The regressions are run on insurance data from SICOR and correspond to equation (24) for each of the three main crops c affected by the frost shock: corn, coffee, and wheat. Outcome is given by $\left(\frac{\text{Premium}}{\text{Insured Value}}\right)$, which is our measure of insurance premium. There is no significant difference on any crops insurance premium between affected and non affected farmers.

B Data Appendix

We combine all electronic payments at the Central Bank of Brazil: boletos, bank transfers, and instant payments. Because boletos can be settled via cash, and it is standard to pay by boleto when issuing a nota fiscal for tax purposes, we believe that we cover almost all of the farmers' input payments. We classify the following CNAE codes as upstream of coffee farming, whose code is 0134-2/00, for the sake of the payment regressions that we interpret as input purchases proxying for investment:

- **0141-5/01**: Production of certified seeds, except forage for pasture.
- **0142-3/00**: Production of certified coffee seedlings, including genetically modified ones.

- **0161-0/02**: Pruning services in coffee plantations.
- **0161-0/03**: Contracted land preparation, cultivation, and harvesting services.
- **2831-3/00**: Manufacture of machinery and equipment for agriculture and livestock, parts and accessories.
- **2832-1/00**: Manufacture of machine tools, parts, and accessories.
- **3314-7/11**: Maintenance and repair of machinery and equipment for agriculture and livestock.
- **3314-7/12**: Maintenance and repair of machine tools.
- **4530-7/05**: Retail trade of tires and inner tubes.
- **4661-3/00**: Wholesale trade of machinery, devices, and equipment for agricultural use; parts and pieces.
- **7731-4/00**: Rental of agricultural machinery and equipment without operator.
- **7739-0/99**: Rental of other machinery and equipment not previously specified, without operator.
- **7490-1/03**: Intermediation and agency activities of services and businesses in general, except real estate.
- **0159-8/99**: Production of other plant-based products not previously specified.
- **0161-0/99**: Support activities for agriculture not previously specified.
- **4611-7/00**: Commercial representatives and trade agents of agricultural raw materials and live animals.

C Model Appendix

C.1 Sequential implementation of the unified collateral constraint

This appendix derives the debt–insurance representation of the unified collateral constraint, presents the household optimality conditions, and describes the numerical and estimation procedures for the model in Section 2.

For a farmer of type a, define harvest revenue and the appraised land value by

$$x_a(S',z';k) \equiv p(S')A_a\theta(S',z')^\alpha k^\alpha, \tag{32}$$
$$q(S',z';k) \equiv \bar{L} + \psi\kappa_G p(S')\theta(S',z')k. \tag{33}$$

We assume that, after default, the lender can recover fraction λ_K of surviving productive capital, fraction λ of harvest revenue, and a claim equal to fraction χ of appraised land value. The maximum enforceable state-contingent claim is therefore

$$\mathcal{B}_a(S',z';k) = \lambda_K\theta(S',z')k + \lambda x_a(S',z';k) + \chi q(S',z';k). \tag{34}$$

The defaulting farmer retains the remaining value and can return to credit markets in subsequent periods. Because land is a fixed endowment in the current model, its appraised value affects the borrowing constraint but not the flow resource constraint. Thus χq is a reduced-form recovery claim rather than an explicit law of motion for land ownership.

Let $d \geqslant 0$ denote one-period non-state-contingent debt and $h(S',z') \geqslant 0$ a state-contingent insurance payoff. Define net repayment as

$$b(S',z') \equiv d - h(S',z'). \tag{35}$$

The debt–insurance constraint and the net-repayment constraint are equivalent:

$$\begin{aligned} d &\leqslant \mathcal{B}_a(S',z';k) + h(S',z'), && \forall(S',z'), \\ b(S',z') &\leqslant \mathcal{B}_a(S',z';k), && \forall(S',z'). \end{aligned} \tag{36}$$

Given any bounded schedule b, set

$$d = \left[\max_{S',z'} b(S',z')\right]^+, \qquad h(S',z') = d - b(S',z'). \tag{37}$$

Then $h \geqslant 0$, condition (36) is exactly $b \leqslant \mathcal{B}_a$, and current net funding is unchanged:

$$\frac{d}{R} - \mathbb{E}\left[\frac{h(S',z')}{R}\right] = \mathbb{E}\left[\frac{b(S',z')}{R}\right]. \tag{38}$$

The reverse direction follows immediately from $b = d - h$. Thus the sequential debt and insurance contracts implement the net-repayment schedule in Proposition 1 for a given enforcement limit.

C.2 Optimality conditions

For a given perceived law of motion for prices, the farmer solves the following recursive problem. We omit the cross-sectional distribution from the notation.

$$V_a(W,S,z) = \max_{k,\{b(S',z')\}} \left\{ u(c) + \beta \sum_{S',z'} \pi(S',z' \mid S,z) V_a(W',S',z') \right\} \tag{39}$$

$$\text{s.t.} \qquad W' = x_a(S',z';k) + \theta(S',z')k - b(S',z'), \tag{40}$$

$$c = W + \sum_{S',z'} \pi(S',z' \mid S,z) \frac{b(S',z')}{R} - k, \tag{41}$$

$$b(S',z') \leqslant \mathcal{B}_a(S',z';k) \quad \forall (S',z'). \tag{42}$$

Let $\xi(S',z') \geqslant 0$ be the multiplier on (42). Let $\mathbb{E}'$ denote expectation over (S',z') conditional on (S,z), let $\pi'_{S'z'} \equiv \pi(S',z' \mid S,z)$, and write

$$x_{a,k}(S',z';k) = \alpha p(S') A_a \theta(S',z')^{\alpha} k^{\alpha-1},$$
$$\mathcal{B}_{a,k}(S',z';k) = \lambda_K \theta(S',z') + \lambda x_{a,k}(S',z';k) + \chi\psi\kappa_G p(S')\theta(S',z').$$

An interior solution satisfies

$$u'(c) = \beta \mathbb{E}'\left[V_{a,W}(W',S',z')\left[x_{a,k}(S',z';k) + \theta(S',z')\right]\right] + \sum_{S',z'} \xi(S',z') \mathcal{B}_{a,k}(S',z';k), \tag{43}$$

$$\beta V_{a,W}(W',S',z') + \frac{\xi(S',z')}{\pi'_{S'z'}} = \frac{u'(c)}{R}, \tag{44}$$

$$V_{a,W}(W,S,z) = u'(c), \tag{45}$$

$$\xi(S',z')\left[\mathcal{B}_a(S',z';k) - b(S',z')\right] = 0. \tag{46}$$

The bond and complementary-slackness conditions hold for every reachable (S', z') with $\pi'_{S'z'} > 0$; zero-probability states are excluded from the choice vector. The expectation in (43) integrates over continuation states. Individual farmers take $p(S')$ as given; the outer equilibrium loop updates it from aggregate output.

When the constraint binds, substituting (34) into (40) gives

$$W_a^B(S', z') = (1-\lambda)p(S')A_a\theta(S', z')^\alpha k^\alpha + \left[1 - \lambda_K - \chi\psi\kappa_G p(S')\right]\theta(S', z')k - \chi\bar{L}. \quad (47)$$

In Regime II-a, $\psi = 0$, so land shifts the level of the limit but not the marginal collateral value of capital. In Regime II-b, surviving capital also creates pledgeable grove value. We impose the normal-state normalization in (7) when comparing the two regimes. Restriction (6) ensures that binding-state wealth remains weakly increasing in surviving capital.

C.3 Numerical solution and validation

The numerical procedure for the full model combines an endogenous-grid household solver with a Krusell–Smith price iteration:

1. For a conjectured aggregate price law and each farmer type, create a grid for next-period wealth and initialize $V_{a,W}$.

2. At each grid point, solve every candidate binding set implied by (43)–(46). Since $p(S')$ and $\theta(S', z')$ both vary across continuation states, we do not impose a fixed best-to-worst ordering. For binding states impose (47); for slack states invert (44). We retain only solutions that satisfy feasibility and complementary slackness in every reachable state.

3. Recover current wealth from (41), update the marginal value using (45), and iterate to convergence. Under land collateral, (47) can approach $-\chi\bar{L}$. We therefore extend the wealth grid below zero but keep admissible wealth above the natural debt limit so that $c > 0$.

4. Simulate the cross-section and compute

$$Y_{t+1} = \int A_i \theta_{i,t+1}^\alpha k_{it}^\alpha \, di, \qquad p_{t+1} = c_p Y_{t+1}^{-1/\gamma}. \quad (48)$$

   Re-estimate the perceived law of motion and repeat until the household policies, price forecasts, and market-clearing residuals converge.

5. Construct a simulated farmer panel with the same outcome definitions as the empirical data and run the same event study code used for the empirical results.

The CPR-only general equilibrium problem is the $\lambda_K = \chi = 0$ case of the recursive characterization above. For the quantitative implementation, we validate the household solver by reproducing the insurance-by-net-worth figure and the debt and investment paths from the solved fixed-price CPR exercise within a stated numerical tolerance. We separately reproduce the capital-constraint policies and longer-horizon results in Appendix C.5. For selected parameter values, we also compare the policy functions and Euler residuals with a coarse-grid value-function solution. We then add land and check nesting across collateral regimes, positive consumption, budget identities, restriction (6), complementary slackness, and market clearing.

C.4 Calibration and Estimation

The benchmark values in Table 1 are inputs rather than estimates. We map the remaining parameters to the data as follows:

1. We benchmark α using external estimates of agricultural production elasticities and output responses to capital, and treat expenditure shares as supporting evidence. We map R from real rural borrowing rates at the model frequency.

2. We map claim and yield losses into (θ_f, δ) using agronomic evidence on physical capital destruction and depreciation, rather than using monetary claim ratios directly.

3. We estimate $\pi(S' \mid S)$ from the historical frost process after using optimal transport quantization to define the discrete model states.

4. We restrict λ using debt relative to pledgeable harvest value among CPR-secured loans, allowing for whether the constraint binds, and use the debt event study path as an additional moment.

5. We estimate the joint distribution of productivity and net worth using pre-shock yield, collateral, credit-to-area, farm size, and insurance take-up.

6. We restrict (γ, c_p) using aggregate coffee production and price data, controlling for demand shifts and using external elasticity estimates as a benchmark.

7. Conditional on validating the SCR guarantee fields, we will restrict χ using land-secured LTVs and the difference in debt responses between the two pre-shock guarantee groups. We will use SCR guarantee values and regional farmland prices to restrict $\bar{L}$, and grove establishment or appraisal values to restrict (ψ, κ_G).

The SMM estimation will match the full event-time paths of debt, capital-replenishment investment, and insurance by pre-shock coverage status, using the same estimator in the empirical and simulated panels. We will also match the difference in the debt response between farmers classified by their predominant pre-shock guarantee. Because the model has one borrowing limit for each farmer, we do not target loan-level collateral shares. National coffee prices are absorbed by the local time fixed effects, so we restrict (γ, c_p) with aggregate price and production data under the stated demand and supply assumptions. The one-period debt model has no direct counterpart for maturity, so we treat the maturity response as an untargeted empirical result until a renegotiation state is added.

The parameter λ_K is active only when reproducing the solved capital-constraint benchmark. We set $\lambda_K = 0$ in the fixed-price CPR exercise and when estimating the CPR and land versions so that productive capital is not counted again through grove value. The main event-study figures come from the solved fixed-price CPR household problem, while the appendix reports the separate capital-constraint benchmark.

C.5 Solved Capital-Constraint Partial-Equilibrium Benchmark

The paper reports two distinct solved fixed-price exercises. Section 2.5 uses the four-node CPR household problem and separates farmers by initial insurance status. This subsection reports the two-state, all-uninsured capital-constraint benchmark. We report its optimal-contract derivation, numerical solution, parameterization, and proposed calibration targets because it provides additional quantitative evidence for the mechanism. To recover its Bellman equation from the unified environment, set $\lambda = \chi = 0$, set λ_K equal to the capital recovery parameter, and take the price process as given. The notation is linked by

$$\widetilde{p}(s') = p(s') A_a \left[\theta^{GE}(s')\right]^{\alpha}, \qquad \theta^{GE}(s') = (1-\delta)\theta^{PE}(s'). \tag{49}$$

The CPR and land versions instead set $\lambda_K = 0$ and make the coffee price consistent with aggregate output in equilibrium.

C.5.1 Optimal Contract

We can write the optimal contract problem as a problem of choosing a sequence $\{c(s^t), t(s^t), k^*(s^t)\}_{t \geqslant \tau}$ to maximize,

$$E_\tau \left[\sum_{t=\tau}^{\infty} \beta^{(t-\tau)} u\left(c_t\right) \right] \tag{50}$$

subject to the current and future period budget constraints:

$$W\left(s^\tau\right) \geqslant c\left(s^\tau\right) + t\left(s^\tau\right) + k^*(s^\tau), \tag{51}$$

$$\tilde{p}(s^t)k^*\left(s^{t-1}\right)^{\alpha}+(1-\delta)\theta(s^t)k^*\left(s^{t-1}\right)\geqslant c\left(s^t\right)+t\left(s^t\right)+k^*(s^t),\quad \forall t>\tau, \tag{52}$$

the lender participation constraint:

$$E_{\tau}\left[\sum_{t=\tau}^{\infty}R^{-(t-\tau)}t_t\right]\geqslant 0 \tag{53}$$

where the transfer plan is restricted so that its discounted value is well-defined. For the debt representation below, we impose the transversality condition

$$\lim_{T\to\infty}R^{-(T+1-\tau)}\mathbb{E}_{\tau}\left[b(s^{T+1})\right]=0. \tag{54}$$

and the limited commitment constraint:

$$E_{\tau'}\left[\sum_{t=\tau'}^{\infty}\beta^{\left(t-\tau'\right)}u\left(c_t\right)\right]\geqslant E_{\tau'}\left[\sum_{t=\tau'}^{\infty}\beta^{\left(t-\tau'\right)}u\left(\hat{c}_t\right)\right],\quad \forall\tau'\geqslant\tau,\forall\{\hat{c}\left(s^t\right)\}, \tag{55}$$

where $\left\{\hat{c}\left(s^t\right)\right\}_{t=\tau'}^{\infty}$ is a solution to the optimal contract when the state variable W is given by

$$\hat{W}\left(s^{\tau'}\right)=\tilde{p}(s^{\tau'})k^*\left(s^{\tau'-1}\right)^{\alpha}+(1-\lambda)(1-\delta)\theta(s^{\tau'})k^*\left(s^{\tau'-1}\right).$$

Endogenous debt limit framework We can write the farmer problem as:

of choosing a sequence $\{c(s^t),b(s^t),k^*(s^t)\}_{t\geqslant\tau}$ to maximize,

$$U\left(\{c(s^t)\}\mid s^{\tau}\right):=E_{\tau}\left[\sum_{t=\tau}^{\infty}\beta^{(t-\tau)}u\left(c_t\right)\right] \tag{56}$$

subject to the sequential budget constraint

$$\begin{aligned}&\tilde{p}(s^t)k^*\left(s^{t-1}\right)^{\alpha}+(1-\delta)\theta(s^t)k^*\left(s^{t-1}\right)+\sum_{s^{t+1}}\frac{1}{R}\pi(s^{t+1}|s^t)b(s^{t+1})\geqslant\\&c\left(s^t\right)+k^*(s^t)+b(s^t),\quad\forall t\geqslant\tau,\end{aligned} \tag{57}$$

and the debt limit is given by a process $\{\widetilde{D}^i\left(s^t\right)\}_{t>\tau}$:

$$\lambda(1-\delta)\theta(s^t)k^*\left(s^{t-1}\right)-b\left(s^t\right)\geqslant-\widetilde{D}^i\left(s^t\right),\quad\forall t>\tau. \tag{58}$$

On the following, we define the continuation utility, conditional on repaying debt equal to x at state s^τ and having debt limit given by $\{\widetilde{D}^i\left(s^t\right)\}_{t>\tau}$ as $\widetilde{V}^i\left(\{\widetilde{D}^i\left(s^t\right)\}_{t>\tau}, x \mid s^\tau\right)$. We say that a sequence of debt limit is *not too-tight* if for all s^t,

$$\widetilde{V}^i\left(\{\widetilde{D}^i\left(s^t\right)\}_{t>\tau}, x \mid s^\tau\right) = \widetilde{V}^i\left(\{\widetilde{D}^i\left(s^t\right)\}_{t>\tau}, 0 \mid s^\tau\right)$$

Following the arguments from Section 3.2 of Martins-Da-Rocha et al. (2022) , we can show that a sequence of debt limits is not too tight if and only if $\widetilde{D}^i\left(s^t\right) = 0$ for all s^t.

Lemma 1. (*Endogenously incomplete markets*) *A consumption allocation is the outcome of the optimal contract if and only if, it is the outcome of the endogenous debt limit framework.*

Proof. ($\Rightarrow$) Fix a sequence $\{c(s^t), t(s^t), k^*(s^t)\}_{t\geqslant\tau}$ that solves the farmer's optimal contract problem. We first show that if a sequence $\{t(s^t)\}_{t\geqslant\tau}$ satisfies (53) and (55), then

$$\lambda(1-\delta)\theta(s^{\tau'})k^*\left(s^{\tau'-1}\right) \geqslant \mathbb{E}_{\tau'}\left[\sum_{t=\tau'}^{\infty} \frac{t_t}{R^{t-\tau'}}\right], \quad \text{for all } \tau' \geqslant \tau. \tag{59}$$

Assume by way of contradiction that there exists $\tau', s^{\tau'}$ such that

$$\lambda(1-\delta)\theta(s^{\tau'})k^*\left(s^{\tau'-1}\right) < \mathbb{E}_{\tau'}\left[\sum_{t=\tau'}^{\infty} \frac{t_t}{R^{t-\tau'}}\right].$$

Consider the following deviation. The farmer defaults at τ' and enters a contract with the following allocation schedule:

$$\hat{t}(s^t) = \begin{cases} t(s^t) & \text{if } t > \tau' \\ -\mathbb{E}_{\tau'}\left[\sum_{t=\tau'+1}^{\infty} \frac{t_t}{R^{t-\tau'}}\right] & \text{if } t = \tau' \end{cases}$$

$$\hat{k}^*(s^t) = k^*(s^t), \quad \text{for all } t \geqslant \tau'.$$

By construction, $\hat{c}(s^t) = c(s^t)$ for all $t > \tau'$. We choose $\hat{c}_{\tau'}$ so that the budget constraint holds at τ'. Moreover, the definition of $\hat{t}$ ensures that (53) holds. The change in consumption at the default date is

$$\begin{aligned}\hat{c}_{\tau'} - c_{\tau'} &= t(s^{\tau'}) - \hat{t}(s^{\tau'}) - \lambda(1-\delta)\theta(s^{\tau'})k^*\left(s^{\tau'-1}\right) \\ &= \mathbb{E}_{\tau'}\left[\sum_{t=\tau'}^{\infty} \frac{t_t}{R^{t-\tau'}}\right] - \lambda(1-\delta)\theta(s^{\tau'})k^*\left(s^{\tau'-1}\right) > 0,\end{aligned}$$

which contradicts optimality. The remainder of the proof is analogous to Rampini and

Viswanathan (2013). Define

$$b\left(s^{\tau'}\right) := \mathbb{E}_{\tau'}\left[\sum_{t=\tau'}^{\infty} R^{-\left(t-\tau'\right)} t_t\right] \leqslant \lambda(1-\delta)\theta(s^{\tau'})k^*\left(s^{\tau'-1}\right), \quad \forall \tau' > \tau.$$

Then $t_{\tau'} = b(s^{\tau'}) - \sum_{s^{\tau'+1}} \frac{1}{R}\pi(s^{\tau'+1}|s^{\tau'})b(s^{\tau'+1})$ for all $\tau' \geqslant \tau$, with $b(\tau) = 0$. Then the budget constraints (57) become equivalent to the budget constraints (51). Moreover, (58) holds, so the sequence of consumption is feasible in the economy with endogenous debt limit.

($\Leftarrow$) Fix a sequence $\{c(s^t), b(s^t), k^*(s^t)\}_{t\geqslant\tau}$ that solves the farmer's maximization problem under the endogenous debt limit framework. Define the repayment values, for all $t > \tau$:

$$t(s^t) := b(s^t) - \sum_{s^{t+1}} \frac{1}{R}\pi(s^{t+1}|s^t)b(s^{t+1}),$$

Consider the sequence $\{t(s^t)\}_{t\geqslant\tau}$. This sequence satisfies the budget constraint (51) by definition. Moreover, (53) is satisfied.[11]

$$\mathbb{E}_\tau\left[\sum_{t=\tau}^{\infty} R^{-(t-\tau)} t_t\right] = 0.$$

Now assume by way of contradiction that $\{t(s^t)\}_{t\geqslant\tau}$ does not satisfy (55). Then there exists $\tau' \geqslant \tau$ such that $U\left(\{c(s^t)\} \mid s^{\tau'}\right) < U\left(\{\hat{c}(s^t)\} \mid s^{\tau'}\right)$, where $\{\hat{c}(s^t)\}$ solves the optimal contract problem with initial net worth $\tilde{p}(s^{\tau'})k^*(s^{\tau'-1})^\alpha + (1-\lambda)(1-\delta)\theta(s^{\tau'})k^*(s^{\tau'-1})$. Let the associated transfers be $\hat{t}(s^t)$. By the previous part of the proof,

$$\hat{b}(s^t) := \mathbb{E}_t\left[\sum_{k=t}^{\infty} R^{-(k-t)}\hat{t}_k\right], \quad \forall t > \tau'$$

is feasible in the endogenous debt limit with $b(s^{\tau'}) = 0$. Now define the alternative con-

[11] For every finite T, iterated expectations give

$$\mathbb{E}_\tau\left[\sum_{t=\tau}^{T} R^{-(t-\tau)} t_t\right] = b(s^\tau) - R^{-(T+1-\tau)}\mathbb{E}_\tau\left[b(s^{T+1})\right].$$

We set $b(s^\tau) = 0$. Taking $T \to \infty$ under condition (54) gives a lender value of zero.

sumption plan,

$$\tilde{c}(s^t) = \begin{cases} c(s^t) & \text{if } t < \tau', \\ \hat{c}(s^t) & \text{if } t \geqslant \tau'. \end{cases}$$

This consumption plan is feasible under the endogenous debt limit problem because at τ' we have $b(s^{\tau'}) \leqslant \lambda(1-\delta)\theta(s^{\tau'})k^*(s^{\tau'-1})$. It follows that $U\left(\{c(s^t)\} \mid s^\tau\right) < U\left(\{\tilde{c}(s^t)\} \mid s^\tau\right)$, contradicting optimality. □

Corollary 1. (*Alternative characterization of endogenously incomplete markets*) *A consumption allocation is the outcome of the optimal contract if and only if the allocation is the outcome of an economy where farmers only have access to a sequence of one-period state contingent savings contracts* $\{h(s^t)\}_{t\geqslant\tau}$ *and one period* (*not state contingent*) *debt contracts* $\{d(s^t)\}_{t\geqslant\tau}$*, satisfying :*

$$\lambda(1-\delta)\theta(s^{\tau'})k^*\left(s^{\tau'-1}\right) + h(s^{\tau'}) \geqslant d\left(s^{\tau'-1}\right), \quad \forall \tau' > \tau \tag{60}$$

$$d(s^t) \geqslant 0, \quad h(s^t) \geqslant 0, \quad \textit{for all } t > \tau$$

Proof. We obtain the result by defining, for every $\tau' > \tau$ and every history $s^{\tau'}$,

$$d(s^{\tau'-1}) := \left[\max_{s^{\tau'}} b(s^{\tau'})\right]^+$$

$$h(s^{\tau'}) := d(s^{\tau'-1}) - b(s^{\tau'}),$$

and appeal to Lemma 1. □

C.5.2 General Solution Technique

Below is an algorithm to solve the problem

$$\begin{aligned} V(W,s) = & \max_{c,k,\{W(s')\},\{b(s')\}} u(c) + \beta\mathbb{E}[V(W(s'),s') \mid s] \\ \text{s.t.} \quad & W + \sum_{s'} \pi(s' \mid s)\frac{b(s')}{R} \geqslant c + k, \\ & \tilde{p}(s')k^\alpha + (1-\delta)\theta(s')k \geqslant W(s') + b(s'), \quad \text{for all } s', \\ & \lambda(1-\delta)\theta(s')k \geqslant b(s'), \quad \text{for all } s'. \end{aligned}$$

Note that we can rewrite this equation as

$$
\begin{aligned}
V(W, s) = \max_{k, \{b(s')\}} u & \left(W + \sum_{s'} \pi(s' \mid s) \frac{b(s')}{R} - k \right) \\
& + \beta \sum_{s'} \pi(s' \mid s) V \left(\tilde{p}(s') k^{\alpha} + (1-\delta)\theta(s') k - b(s'), s' \right) \\
\text{s.t.} \quad & \lambda(1-\delta)\theta(s') k \geqslant b(s'), \quad \text{for all } s'.
\end{aligned}
$$

For compactness, define

$$
q(s') \equiv (1-\delta)\theta(s'), \qquad Y(s'; k) \equiv \tilde{p}(s') k^{\alpha} + q(s') k.
$$

The algorithm is as follows:

- **Inputs:** Parameters, a grid $\mathcal{W} = (W_l, \ldots, W_h)$ for next-period wealth, an initial guess for $V_W(W, s)$, and an interpolation rule outside the grid.
- In the unconstrained case, capital satisfies

  $$
  k^*(s) = \left(\frac{\alpha \sum_{s'} \pi(s' \mid s) \tilde{p}(s')}{R - \sum_{s'} \pi(s' \mid s) q(s')} \right)^{\frac{1}{1-\alpha}}.
  $$

  When no continuation-state constraint binds, the marginal-utility ratio is degenerate at $k^*(s)$. We recover $u'(c)$ from $u'(c) = \beta R V_W(W(s_j), s_j)$ at the grid anchor and then recover wealth in the remaining states. For a binding anchor state s_j and a grid value $W(s_j) = w_g$, recover k from

  $$
  w_g = \tilde{p}(s_j) k^{\alpha} + (1-\lambda) q(s_j) k.
  $$

- For each current state s, grid value w_g, and candidate set $\mathcal{B}$ of binding continuation-state constraints:
  1. For each $s_j \in \mathcal{B}$, impose

     $$
     W(s_j) = Y(s_j; k) - \lambda q(s_j) k = \tilde{p}(s_j) k^{\alpha} + (1-\lambda) q(s_j) k.
     $$

  2. Compute marginal utility from

     $$
     u'(c) = \frac{\beta R \sum_{s_j \in \mathcal{B}} \pi(s_j \mid s) V_W(W(s_j), s_j) \left[\alpha \tilde{p}(s_j) k^{\alpha-1} + (1-\lambda) q(s_j) \right]}{R - \sum_{s_j \in \mathcal{B}} \pi(s_j \mid s) \lambda q(s_j) - \sum_{s_j \notin \mathcal{B}} \pi(s_j \mid s) \left[\alpha \tilde{p}(s_j) k^{\alpha-1} + q(s_j) \right]}.
     $$

Recover c by inverting u'.

3. For each $s_j \notin \mathcal{B}$, find $W(s_j)$ such that

$$\beta R V_W(W(s_j), s_j) = u'(c).$$

4. Recover repayment in every state from

$$b(s_j) = Y(s_j; k) - W(s_j).$$

Retain the candidate only if $b(s_j) = \lambda q(s_j)k$ for binding states and $b(s_j) < \lambda q(s_j)k$ for slack states.

5. Recover current wealth from

$$W = c + k - \sum_j \frac{\pi(s_j \mid s)b(s_j)}{R},$$

and update $V_W^{\text{next}}(W, s) = u'(c)$.

- Update V_W and repeat until $\|V_W - V_W^{\text{next}}\|_{\infty,\mathcal{W}}$ is below tolerance.

C.5.3 Calibration Targets

For the quantitative extension, we use the following mappings from parameters to empirical moments. These mappings are not needed to generate the partial-equilibrium figures reported here. Table 4 lists the parameter values actually used in the solved simulation.

1. Cobb-Douglas capital parameter α : Farmer's expenditure on capital (interest payments on debt + outbound payments in the specified CNAE codes that we interpret as investment) as a fraction of total expenditure (interest payment + RAIS wages if we have that + all outbound payments)

2. R : Average (real) borrowing interest rate.

3. Demand elasticity γ : Coffee price elasticity with respect to Brazil's coffee production

$$\gamma = \frac{\gamma_w}{s_B},$$

where γ_w is the world price elasticity with respect to total quantity, and s_B is the average share of Brazilian coffee exports.

4. Shock magnitude and transition probabilities θ, π : Optimal transport based quantization.

5. Borrowing constraint parameter λ : Iterate over model to match the debt event studies.

6. Distribution of productivity $A_{i,t}$: Two types A_{un} and A_{ins}. Set $A_{un} = 1$ and A_{ins} to match the share of insured and uninsured in the data.

7. Discount rate β and CRRA risk aversion σ: See below.

Let $a_i(s_i^t)$ denote the sum of the value of the capital stock $k_i(s_i^t)$ and income net of the insurance claim $y(s_i^t)$. We jointly calibrate σ and β using these two moments: 1) unconstrained (insured) farmers' consumption response to the net worth shock inclusive of insurance payout

$$\theta_a = \frac{\partial \Delta \log c}{\partial \Delta \log a}$$

and 2) their steady-state ratio of consumption to income (1− savings rate). If the event study is not identified or there are other objections, use the ratio $\rho := \frac{\text{Var}(\Delta \log c)}{\text{Var}(\Delta \log y)} = (1 - \theta_a)^2$. We proxy consumption using outbound payments to CNAE codes that are purchases of final goods, defined as the purchase share to CPFs (vs CNPJs) being above a threshold. We proxy income using inbound boletos, scaled by the municipality-level data on coffee production multiplied by the coffee price.

We do not use a closed-form mapping from these moments to (σ, β) in the reported results. The quantitative estimation will solve this mapping numerically.

C.5.4 Numerical Simulation

We consider a continuum of farmers with unit mass. Each farmer receives an idiosyncratic shock to capital survival, which takes values $\theta = 0.3$ in the bad state and $\theta = 0.95$ in the good state. The benchmark features two aggregate states. In the good aggregate state, 20% of farmers receive the bad idiosyncratic shock. In the bad aggregate state, 50% of farmers receive the bad idiosyncratic shock.

We use $\alpha = 0.3$, $\lambda = 0.7$, $\beta = 0.9$, $R = 1.05$, and CRRA utility with $\sigma = 2$. Within this solved benchmark, λ is the capital recovery parameter denoted by λ_K in the unified constraint. The transition matrix is set so that the unconditional distribution over aggregate states is

$$P_S = [0.8,\ 0.2]. \tag{61}$$

Table 4: Parameters in the Solved Partial-Equilibrium Benchmark

Parameter	Value	Interpretation
α	0.30	Capital elasticity
λ_K	0.70	Recoverable share of productive capital
β	0.90	Discount factor
R	1.05	Gross funding rate
σ	2.00	CRRA coefficient
θ_f	0.30	Capital survival in the bad state
θ_n	0.95	Capital survival in the good state

The simulation design uses 11,000 periods with 10,000 farmers and discards the first 1,000 periods. Define $\text{Shock}_{it} = 1$ if the aggregate state is bad in period t and farmer i receives the idiosyncratic shock. The simulated-panel regression is

$$y_{it} = \sum_{\tau=-G}^{M} \beta_{\tau}^{\text{NI}} \, \text{Shock}_{i\tau} + \alpha_i + \alpha_t + \varepsilon_{it}, \tag{62}$$

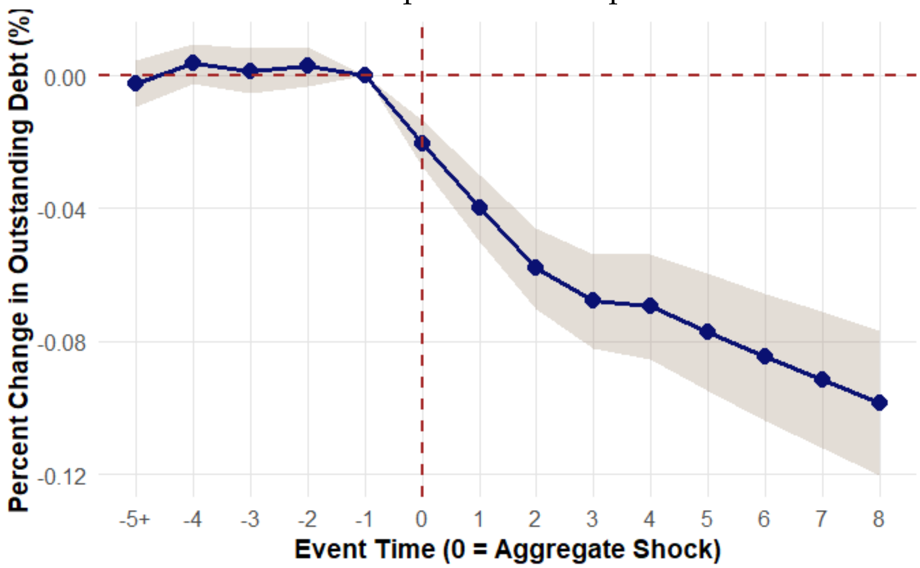
where all farmers in this simulation are uninsured. Section 2.5 reports the separate solved fixed-price CPR exercise with distinct initial insurance groups. The full long-run simulation with both CPR and land collateral is the quantitative extension described in Section 2.4.

C.5.5 Additional Numerical Results

Section 2.5 reports the insured and uninsured responses from the fixed-price CPR exercise used to motivate the empirical specifications. For completeness, Figures C1, C2, and C3 report the longer-horizon results from the two-state capital-constraint benchmark. In this simulation, all farmers in the event-study regression are uninsured.

We compute the cumulative responses of farmers' debt load to the shock using the model-implied event-study coefficients. Figure C1 shows a persistent decline in borrowing following the shock. The frost reduces the capital that the farmer can pledge, and the borrowing constraint remains tight during the recovery.

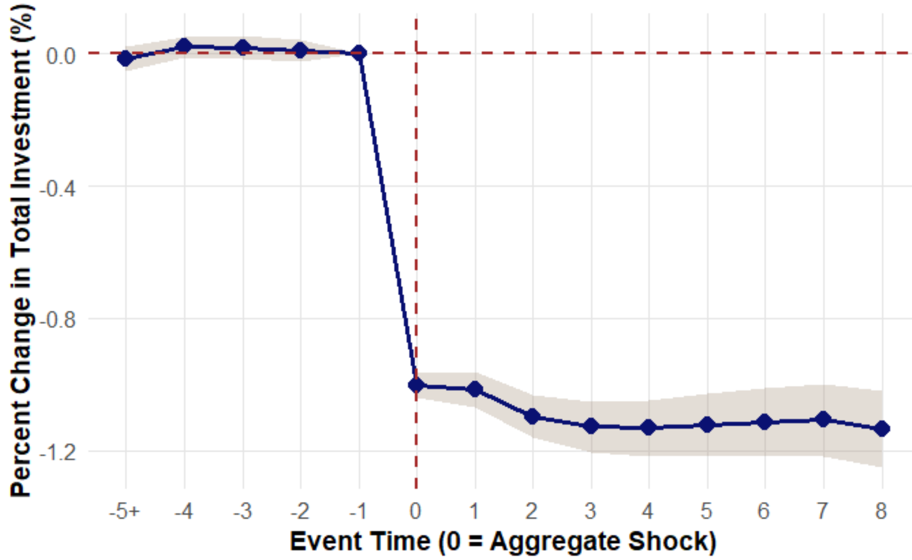
Figure C1: Cumulative Model-Implied Debt Response to an Adverse Shock



Notes: Based on model-simulated data, we estimate the cumulative effect of an adverse shock on debt using a fixed-effects event study. The estimated coefficients capture the dynamic adjustment in debt relative to the quarter of the shock. All farmers in this simulation are uninsured.

Figure C2 shows a prolonged decline in investment following the shock. This pattern is the opposite of what one would expect under complete markets, because the farmers affected by the shock have a higher marginal product of capital and should invest more to replenish their capital stock. Under limited pledgeability, the loss of capital tightens the constraint and prevents this adjustment.

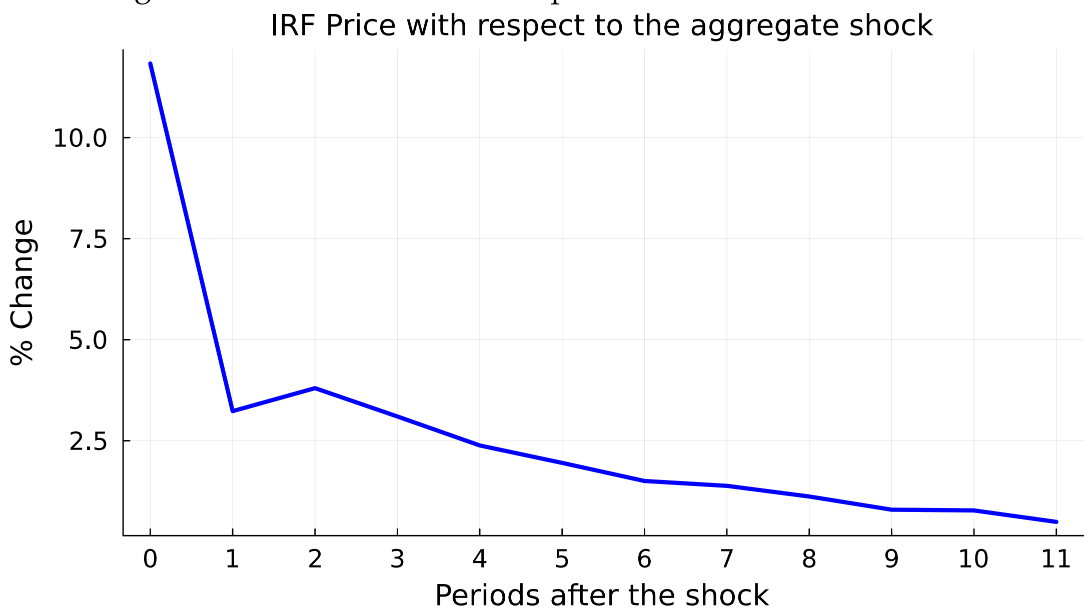
Figure C2: Cumulative Model-Implied Investment Response to an Adverse Shock



Notes: The plot shows the dynamic response of investment following a negative shock, based on the same simulated economy used for the debt results. The specification mirrors that of the debt response, with farmer and time fixed effects.

The imposed price path partially mitigates the revenue and net-worth loss for farmers with more surviving production. It raises internal funds but does not directly relax the capital-based borrowing limit. The direct collateral-price effect arises only after CPR and grove value enter the constraint. Figure C3 preserves the path used in the original benchmark. In other words, it is not an equilibrium response of the partial-equilibrium model.

Figure C3: Coffee-Price Path Imposed after an Adverse Shock



Notes: The plot displays the coffee-price path imposed following a negative shock to productive capital in the solved partial-equilibrium benchmark. It is not an equilibrium response in this exercise.

C.6 CPR-Only General Equilibrium Quantification Plan

The CPR-only general equilibrium model sets $\lambda_K = \chi = 0$ and uses the output-based borrowing constraint. Its quantitative implementation will combine the endogenous grid method with a Krusell–Smith price iteration. We will approximate the law of motion for aggregate output using moments m that summarize the distribution of $A_i k_{it}^{\alpha}$. Farmers will form expectations over next period's output using

$$Y_{t+1}(S_{t+1}) = \int_0^1 A_i \theta(s_{i,t+1})^{\alpha} k_{i,t}^{\alpha}\, di = f(m, S_t, S_{t+1}; a), \tag{63}$$

where a is a finite-dimensional vector that summarizes beliefs. Starting with an initial guess a_0, the algorithm will solve the farmer's problem, simulate the economy, update the belief parameters, and determine the coffee price from inverse demand. These steps will repeat until the price forecast and market-clearing residuals converge. This quantification is separate from the solved partial-equilibrium figures above.